Alpha↔Omega Dynamics (AΩD)

The Hidden Temporal Dynamics of Stokes

The Art Of The Leprechaun: Fractal Calculus – $\mathfrak{S}$

43 °c

Manual Volume II — Matter I: Elements 1–118

Relational reference, isotope catalog and declared survival application

Reginald Poyau

November 2026

$$A \leftrightarrow_\mu \Omega \longleftrightarrow \Omega \leftrightarrow_\mu A$$

Epitaph

$$\mathcal{N} \equiv \infty \qquad \mathcal{N} \to \mathcal{N} \to \mathcal{N} \equiv \mathcal{N}$$

Comprehension, cooperation, and presence are not prerequisites of AFC.

Nuff said.

Contents

1 Manual Volume II Scope and Record Discipline . . . 3
1.1 Data support and record classes . . . 3
1.2 Reading route and evidence scope . . . 4
1.3 Exact arithmetic and external-comparison quarantine . . . 4
2 Imported Relations and Conditional Support Arithmetic . . . 4
2.1 Distinction, motif and instance evidence . . . 4
2.2 Support polynomial and the reflection-duration domain . . . 5
3 Enclosure Membership and the Visual Model . . . 6
3.1 Even enclosure sequence and cumulative boundaries . . . 6
3.2 Relations before layout . . . 7
4 Conditional Support Expressions and Assigned Inputs . . . 9
4.1 Inherited fusion domain and the present calculation . . . 9
4.2 Assigned classifications, residuals and occupancy . . . 10
4.3 Complete reference ledger . . . 10
5 Verification Scope and Pending Native Evidence . . . 11
5.1 Executed static checks and inherited audit obligations . . . 11
6 Folded Construction and Relational Time . . . 12
6.1 Distinction, handed hinge and retained lineage . . . 12
6.2 An injective two-face example . . . 12
6.3 Declared recurrence and mapped comparison . . . 13
6.4 Causal returned-current response and exposure . . . 14
7 Hyperfine Target Records and Observation Map . . . 15
7.1 Atomic identity and sourced recurrence . . . 15
7.2 Hydrogen-referenced deuterium and tritium data . . . 16
7.3 A common model and its atomic observation map . . . 17
8 State-Dependent Evolution and SI Report Maps . . . 18
8.1 A response that changes later events . . . 19
8.2 The atomic observable is an additional construction . . . 19
8.3 Elapsed seconds and a joint uncertainty budget . . . 21
8.4 Source-bound metrological comparisons . . . 22
8.5 Published frequency covariance . . . 23
9 Isotope Half-Lives and a Common Survival Experiment . . . 23
9.1 Source records and nucleon bookkeeping . . . 24
9.2 Selected implantation-to-decay records . . . 25
9.3 A state-dependent finite law . . . 25
9.4 A measured test of the restricted specialization . . . 27
9.5 Constructed-field cadence . . . 28

9.6 Witnessed response, burden and retention . . . 29
9.7 Two configurations under one common law . . . 30
10 An All-Element Inner-Recurrence Campaign . . . 31
11 State-Resolved Element Properties . . . 33
12 Calculation and Data Registry . . . 35
A Exact Arithmetic Fixtures . . . 37
A.1 Scope . . . 37
A.2 Four-step decompositions . . . 37
A.3 Labelled-reference occupancy . . . 37
A.4 Falsification . . . 37
B Quarantined External Comparison . . . 38
B.1 Target records and numeric policy . . . 38
B.2 Observable-map comparisons . . . 38
C Exact Parity, Role and Occupancy Comparators . . . 39
C.1 Integer comparators . . . 39
C.2 Signed-magnitude record . . . 39
D Complete Element Reference Ledger . . . 39

1 Manual Volume II Scope and Record Discipline

This volume supplies a compact relational reference for **Matter I: Elements 1–118**, with eight unfilled reference slots completing the declared grouping through slot 126, and a separately scoped folding and recurrence application. It consumes AOD Main mathematics, the M3 observation and comparison interfaces, and the bound AFC v2.4 reference. Main owns the ontology, hypotheses and mathematics; this Manual owns the elemental inputs, worked constructions, declared update model, visual model, validation and historical crosswalk.

1.1 Data support and record classes

Declare the input distinctions and their induced relations before a calculation, report map or residual. The simulation data card inherited from the bound AOD Manual is

$$(D, C_{\text{support}}, M_{A\leftrightarrow\Omega}, \Pi_{\text{report}}, \sigma_{\text{input}}, \epsilon_{\text{report}}, \text{src}). \tag{1}$$

Here D names the input data, C_{support} its support class, $M_{A\leftrightarrow\Omega}$ the calculation or map, Π_{report} the reported coordinate, σ_{input} the input uncertainty where applicable, ϵ_{report} the residual coordinate, and src the source record. Exact arithmetic of an input does not determine its physical uncertainty. A source record can be declared before a model-to-target map exists.

Table 1: **Applicable Manual data-support classes.** A class describes the supplied distinctions; it is not an execution or agreement verdict.

Class	Name	Declared support and relation	Report coordinate
D0	Exact internal fixture	Integer gates, channels and comparator bins; internal returned-current relations	Exact integer residual and δ_3 tally
O0	Observable-map fixture	Declared observable-map value and measured-sector target; external comparison map	Target residual in the declared coordinate

Record form, scientific claim and data support have distinct roles. An exact fixture stores its comparator and audit. A measured target record stores a quantity, uncertainty, coordinate and citation; a calibration anchor stores a convention value. An external comparison additionally binds the model's map value and target residual. A benchmark manifest declares the required data and report coordinate. Source-only acquisition records do not acquire an executed O0 comparison merely by supplying a target. Static grouping arithmetic is not automatically a D0 returned-current fixture.

The scientific record classes are native exact fixture, declared model/simulation, metrology record and empirical comparison. Native fixtures establish typed relations; models declare update and coupling laws; metrology adds observation, calibration and units; empirical comparison adds sourced targets and residuals. Each application states which of these operations it performs. Admission, execution and empirical agreement remain separate evidence claims. The G0–G3 and L0–L3 support classes of Manual I concern its declared field-dynamics and lensing data; they are not additional Matter I maturity levels.

Provenance. AOD Manual I at commit `71cbcf7`: `manual/sections/00_scope.tex`, labels `manual:scoped-errors-data-support` and `manual:record-forms`; `manual/sections/06_field_dynamics_applications.tex`, `tab:manual-data-support-classes`. The scientific and authoring

bindings are recorded in `manual2/evidence/campaign-source-binding.json`. The copied Manual I source records are bound in `manual2/evidence/data-classes/source-bindings.json`; the component-level cards are in `manual2/data/m1/classification/data_cards.json`.

1.2 Reading route and evidence scope

The dependency order is axiom, distinction, induced relations, admitted motif, declared model, mathematical consequence. The inherited route is Main §1, Main §2, Main §3, Main §4, and Main §5. The present visual model projects declared enclosure membership and local roles. It does not instantiate the native construction trace for each element.

The active element table records exact grouping and support arithmetic, assigned fixture inputs and their consequences. Its support declarations are formal syntax, distinct from cycle or fusion certificates. D.E.C. denotes the Declared Edge Computation ledger: it records a declared raw AFC calculation, admissible edges, exact kernel, route and boundary audit. The ledger stores evaluated values; it does not replace the calculation or join measured targets into it. The finite construction in §6 and the all-element campaign in §10 bind their own events, scope maps, declared laws and returned-current comparisons. Their evidence does not change the class of the static reference table.

1.3 Exact arithmetic and external-comparison quarantine

The support polynomial, cumulative boundaries, local counts, conditional support sums and assigned-input residuals use exact integers. Labelled-slot occupancy and the model's temporal comparisons use exact rationals. A measured value retains its quantity, unit, uncertainty, source and rounding even when its transcribed decimal is evaluated as an exact fraction. A native result precedes its external observation/value map and target join.

The hydrogen/caesium inventory in §7 supplies measured target and calibration records. §8 separates state-dependent evolution, SI maps, a conventional benchmark and observation-law countermodels. §9 separates the sourced isotope catalog from the declared survival ensemble. Its reference-return count and an SI elapsed interval have different report coordinates. App. B defines the external records and comparison map; §12 lists the calculation and data forms.

Provenance. The paired Main publication is built from AOD commit `71cbcf7` before this Manual. Earlier foundation sources and source-dependent executors are archived for the frozen application records. Imported equations and sections resolve through its semantic labels. Historical v94 and v40.03r24 records are comparison inputs, with their original statuses preserved; they do not replace this foundation.

Literature note. Proof discipline and released foundation: Poyau [17, 18, 19].

2 Imported Relations and Conditional Support Arithmetic

2.1 Distinction, motif and instance evidence

Setup. Main distinguishes Null sameness from committed finite events (Main (1), Main (3)). The branch/hinge/counterbranch motif is

$$M_{\varepsilon,m} = \varepsilon\, 0_H (0_D)^m (-\varepsilon),$$

with mandatory hinge and optional dwell (Main (5)). A faithful atomic edge changes one bit of Q_4 (Main (8)). These relations provide the motif template used by the visual projection.

Derivation. For a native instance, a declaration must bind its scope, lineage and construction:

$$\mathbf{a} \vdash (F, R, \partial_T R, \omega), \qquad B = (\partial_T F, \omega). \tag{2}$$

A certified closure additionally binds the retained-start return witness and at least two committed phase-bearing traversals (Main (13), Main (15)). The admitted motif record in Main (33) and traced admission in Main (37) are prerequisites, not consequences of writing an elemental label.

Specialization. This reference supplies grouping, local tuples and conditional expressions for labelled rows. It does not supply the scope-bound instance evidence in (2). Names identify the inherited distinctions or the declared reference records; they introduce no additional object or proof condition. Conversely, names cannot discharge the hypotheses of the construction they identify.

Falsification. A formatted support declaration represented as a successful certificate, or a plotted line represented as an executed edge without its trace, exceeds this profile.

Provenance. Main (5), Main (13), Main (33), Main §5.17.10.

2.2 Support polynomial and the reflection-duration domain

Setup. A support expression $\Sigma = (s_1, \ldots, s_k; L)$, with clauses $s_i = p_i{:}q_i$, is formal syntax (Main (36)). The declared finite support model is

$$\Pi(p{:}q) = q^p + q, \qquad p, q \in \mathbb{Z}_{\geq 1}. \tag{3}$$

Its provenance is Main (35) and Main (174). Admission of the proposed form remains

$$K_{\mathrm{prop}} \xrightarrow{HT(K;B)} K_{\mathrm{adm}}(B). \tag{4}$$

Derivation. For an admitted instance in the domain of the inherited accessor,

$$RD_\gamma(B) = \Pi_\gamma^-(B) + H_{\mu,\gamma}(B) + \Pi_\gamma^+(B). \tag{5}$$

Under equal branch supports and $H_{\mu,\gamma} = 1$,

$$RD_\gamma^{(1)}(B) = 2\Pi(p{:}q) + 1. \tag{6}$$

These are Main (31), Main (32) and Main (176). Evaluating the right-hand expression numerically does not establish its accessor prerequisites.

Specialization. Table 2 preserves the exact polynomial values. Its last column is the conditional single-hinge expression; it is not evidence that the corresponding elemental instances have been admitted.

Table 2: **Imported support polynomial and conditional single-hinge expression.**

Clause	q^p	$\Pi = q^p + q$	$2\Pi + 1$
1:2	2	4	9
3:1	1	2	5
3:2	8	10	21
3:3	27	30	61
3:4	64	68	137

The ordered count basis is

$$\mathcal{B}_{\mathrm{curl}} = (3{:}3,\ 3{:}4,\ 1{:}2), \tag{7}$$

and its local declaration expands as

$$a{:}b{:}c :: L \equiv (3{:}3)^a :: (3{:}4)^b :: (1{:}2)^c :: L. \tag{8}$$

These local counts are distinct from the nuclear counts $(Z, A-Z)$ in §4. Main's type distinctions remain unchanged: Dimonon = 2 Monon $\neq$ Dion, and tetrion, tetron, tetratrion and tetrad denote different constructions.

Falsification. The single-hinge expression cannot replace (5) for unequal branches or a non-unit hinge. A declaration string cannot supply a missing construction witness or cycle-valued compression.

Provenance. Main App. B.4.1, Main App. B.1.10, Main (34), Main (40), Main (43).

3 Enclosure Membership and the Visual Model

3.1 Even enclosure sequence and cumulative boundaries

Setup. The inherited enclosure grammar distinguishes E_2 (duon), its open and seated states $E_2^0, E_2^{\pm}$, the inner E_4 tetron, and outer even support enclosures C_{2m}, $m \geq 3$. The present chart uses C_6 through C_{22} as nested enclosure interfaces. Their displayed geometry represents declared enclosure membership; no measured radius or metric scale is assigned. The semantic sources are Main §5.7 and Main App. B.3.

Derivation. The coord-0 declaration and filled boundary form are

$$0 \mid - :: Z : A : 0 : L\langle a : b : c :: L\rangle, \tag{9}$$

$$0 \mid - :: Z : A : 0 : 2m\langle m : m : 0 :: 2m\rangle. \tag{10}$$

Following Main §5.7.13,

$$\Gamma_m = \sum_{k=3}^{m} 2k = m(m+1) - 6, \qquad \Gamma_2 = 0, \tag{11}$$

and

$$\Gamma_{m-1} < Z \leq \Gamma_m, \qquad s = Z - \Gamma_{m-1} \tag{12}$$

select the unique group and seat. Here $Z = 1, \ldots, 118$ identifies the labelled elements; integers 119–126 are unfilled reference-slot indices in the same grouping.

The local tuple is

$$a(s) = \lceil s/2 \rceil , \qquad b(s) = \lfloor s/2 \rfloor , \qquad c(s) = s \bmod 2. \tag{13}$$

Thus $c = 1$ records odd local parity and $c = 0$ even local parity. It does not determine an assigned acceptor, neutral or shedding class. For example, Li has $s = 3, c = 1$ but an assigned neutral class; O has $s = 2, c = 0$ but an assigned acceptor class. The tuple and those input classifications are distinct relations.

Specialization. The nine groups in Table 3 contain 126 reference slots, of which 118 carry element labels:

$$118/126 = 59/63, \qquad N_{\text{labelled}}(\mathsf{C}_{22})/22 = 14/22 = 7/11.$$

These are labelled-reference occupancy fractions. No retention duration or physical limit beyond 118 follows from them.

Table 3: **Declared enclosure groups and local role positions.** The last endpoint is an unfilled reference slot.

m	$2m$	Γ_{m-1}	Γ_m	Index range	Hinge $s = m$	Boundary $s = 2m$
3	6	0	6	1–6	3 Li	6 C
4	8	6	14	7–14	10 Ne	14 Si
5	10	14	24	15–24	19 K	24 Cr
6	12	24	36	25–36	30 Zn	36 Kr
7	14	36	50	37–50	43 Tc	50 Sn
8	16	50	66	51–66	58 Ce	66 Dy
9	18	66	84	67–84	75 Re	84 Po
10	20	84	104	85–104	94 Pu	104 Rf
11	22	104	126	105–126	115 Mc	126 –

3.2 Relations before layout

Setup. The visual model preserves $Z \mapsto (m, s, a, b, c)$. In each enclosure, $1 \le s < m$ projects the first branch, $s = m$ the hinge, $m < s < 2m$ the counterbranch, and $s = 2m$ the filled-tuple boundary. This projection does not execute inter-element edges.

Derivation. Nested outlines express enclosure membership; labels express local order and role. A spiral can project seat order while preserving membership, but its curvature supplies neither a temporal trajectory nor a transition witness. Assigned classes retain their input meaning from §4.

Falsification. Changing preserved Z, m, s, equating odd parity with acceptor status, or treating unfilled slots as physical non-retention contradicts this model. Field count $n_F \log n_F$ in Main (91) has a separate scope; ring counts cannot supply its field-family witness.

Provenance. Main §5.7.10, Main §5.7.13, Main §5.8, Main §5.7.12, (12), (13).

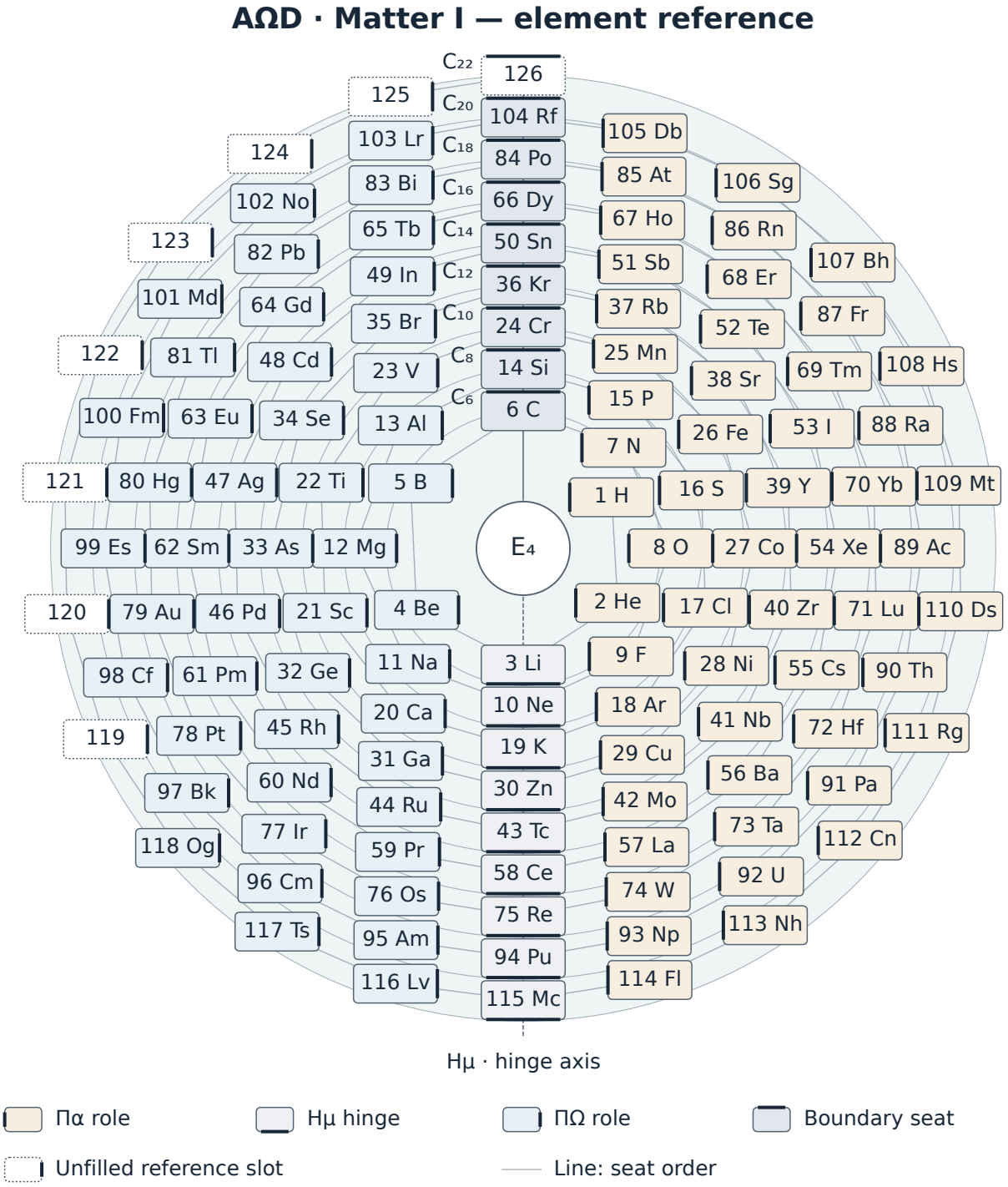


Figure 1: **Matter I static relational reference.** Nested enclosure membership and repeated branch/hinge/counterbranch roles organize all 118 element labels. The line is a seat-order guide. Positions 119–126 are unfilled reference slots in the declared grouping. Geometry supplies neither physical scale nor an executed network.

4 Conditional Support Expressions and Assigned Inputs

4.1 Inherited fusion domain and the present calculation

Setup. For admitted forms F, G with certificate-bound cycle-valued compressions $\widehat{F}, \widehat{G}$, Main defines

$$\mathrm{Fuse}_B(F, G) = \mathrm{Close}_B(\widehat{F}\,\mu\,\widehat{G}) \tag{14}$$

and

$$RD_{\mathrm{fuse}}(F, G; B) = \Pi^{-}_{\widehat{F}}(B) + H_{\mu,\widehat{F},\widehat{G}}(B) + \Pi^{+}_{\widehat{G}}(B). \tag{15}$$

Their exact predecessors are Main (40), Main (114) and Main (115). These operators retain their domains. The present elemental table supplies no inner or outer fusion certificate and does not assert an instance of (14).

Derivation. The declared reference operands use the exact polynomial values in Table 2:

$$u_{-} = 30Z + 68(A - Z), \qquad h = 1, \qquad u_{+} = 4s. \tag{16}$$

Define their conditional support sum

$$U(Z, A, s) = u_{-} + h + u_{+} = 30Z + 68(A - Z) + 1 + 4s. \tag{17}$$

This preserves the scalar expression formerly stored as `rd_fuse`. The active field `support_sum` names its established arithmetic scope. Equality to a native reflection-duration accessor additionally requires the inherited instance prerequisites.

Specialization. The nuclear count pair $(Z, A - Z)$ in (16) is distinct from the local tuple (a, b, c) in (13). At fixed Z, changing the supplied integer A leaves m, s, a, b, c unchanged and gives

$$U(Z, A + \Delta A, s) - U(Z, A, s) = 68\Delta A.$$

For carbon, $Z = s = 6$, the conditional choices $A = 12, 13, 14$ give $U = 613,681,749$. This is an algebraic illustration, not a sourced isotope catalog, stability statement or decay/fusion transition.

The reference A values are carried from the reviewed candidate's input list without alteration. Their original per-element selection policy was not recorded. The input provenance identifies that candidate source and this limitation; no broad AME or NIST citation establishes each selected A. Reference A is an integer nucleon-count input, not standard atomic weight. Projected A values for slots 119–126 remain explicitly illustrative fixture operands, not observed nuclide records. The inputs are recorded in `manual2/data/m1/reference_inputs.csv`, `model_inputs.json` and `input_provenance.json` in the same directory.

Falsification. A change in A that changes the grouping or local tuple violates this specialization. Calling a numerically correct U a certified fusion duration without its input and outer certificates exceeds its domain.

Provenance. Main (35), Main (115), (16), (13).

4.2 Assigned classifications, residuals and occupancy

Setup. The fixture keeps the candidate's assignment sets:

$$\mathcal{A} = \{1, 8, 9, 16, 17, 34, 35, 52, 53\}, \qquad \mathcal{X} = \{43, 61\} \cup \{84, \ldots, 118\}.$$

They define assigned acceptor and assigned shedding classes. They are input classifications; they are not derived from Z, A, parity, or an executed native calculation. Let $L = 2m$, $C^{\mathrm{fix}}_{\mathrm{close}} = 3L$, and $w^{\mathrm{fix}} = L$.

Derivation. With assigned pressure P^{fix}, retain the residual arithmetic

$$S^{\mathrm{fix}} = P^{\mathrm{fix}} - C^{\mathrm{fix}}_{\mathrm{close}}, \qquad X^{\mathrm{fix}} = \max(0, S^{\mathrm{fix}}). \tag{18}$$

The pressure input is $C^{\mathrm{fix}}_{\mathrm{close}} - 3$ on $\mathcal{A}$, equal to $C^{\mathrm{fix}}_{\mathrm{close}}$ on the remaining labelled neutral rows, and $C^{\mathrm{fix}}_{\mathrm{close}} + 3u_Z$ on $\mathcal{X}$, where

$$u_Z = 1 \ \ (Z = 43, 61), \qquad u_Z = \lfloor (Z - 84)/6 \rfloor + 1 \ \ (84 \le Z \le 118).$$

For illustrative unfilled slots it is $C^{\mathrm{fix}}_{\mathrm{close}} + 3(Z - 118)$. Assigned duration values are respectively L, $L+2$, $L-1$, and 0. Assigned route and Yro labels are carried as fixture data. These values do not demonstrate retention, instability, transfer or non-retention.

The arithmetic uses the form of Main (107) and Main (108); it does not supply their native pressure, field and window evidence. Likewise, a declared route label does not execute the sheddic partition, lift or higher-scope fusion in Main (110), Main (119) and Main (121).

Specialization. The 118 labelled rows have disjoint assigned residual classes: 72 neutral, 9 deficit/acceptor and 37 positive-surplus/shedding. Boundary position is a separate attribute: eight labelled endpoints are $\{6, 14, 24, 36, 50, 66, 84, 104\}$. Po and Rf are both endpoints and assigned shedding rows. Adding endpoint counts to route-class counts would double-count those attributes. Of the 72 neutral rows, six are endpoints and 66 are not.

The C_{22} fraction 7/11 records 14 labelled slots out of 22. Slots 119–126 are unfilled in this reference; their assigned fixture values establish neither a physical cutoff nor an impossibility of further elements.

Falsification. The assignment table and its equations must agree exactly. Positive X^{fix} for negative S^{fix}, or a changed assigned set without a new input binding, fails the fixture. Agreement with these definitions is not empirical validation.

Provenance. Main (107), Main (108), Main §5.7.12, Main §5.6.6.

4.3 Complete reference ledger

The authoritative generated ledger is `manual2/data/m1/matter1_elements_1_118_ledger.csv`. It binds each static declaration to its input record, distinguishes labelled elements from unfilled slots, and carries pending admission and unexecuted native status. The support declaration is syntax; the support sum and assigned fields retain the scopes above. Assigned route codes are `neutral`, `deficit`, `surplus` and `unfilled`; group-boundary and named-anchor flags are independent attributes. The generated table and figure consume the same validated data. Declared compatibility copies must compare equal.

The complete 118-row reference ledger is retained in App. D; the compact chart remains the primary overview.

5 Verification Scope and Pending Native Evidence

5.1 Executed static checks and inherited audit obligations

Setup. The static checker verifies stored inputs and generated artifacts without rewriting them. Generation is a separate explicit operation. Exact grouping, local tuples, support sums, assignment residuals, identities and data/figure agreement are testable here. Their success does not certify the six native hooks in Main §5.17.

Derivation. Table 4 separates the available static checks from the instance evidence required for native claims. A check result is reported only after that check executes; this table describes scope, not an unrecorded run count.

Table 4: **Verification scope.** Element-wide native obligations remain pending.

Inherited hook	Static reference check	Native evidence not supplied
Spine	Declared role mapping and operand sum	Retained occurrences at each finite depth
Seat/enclosure	Nine boundaries, tuples and labelled occupancy	Scoped field-family and routing witness
Sheddic path	Assigned class, route and residual consistency	Realized path and retention window
Cycle	Support syntax and pending status	Construction, retained start and return certificate
Fusion	Conditional operand and residual equations	Admitted inner forms and outer closure certificate
Naming	Stable declarations, record kinds and input bindings	No name substitutes for instance evidence

Specialization. A future native D.E.C. record has the inherited form

$$(B,\ v,\ e,\ v_e,\ \sigma(e),\ \mathrm{adm}(e;B),\ w(e;B),\ P(e;B),\ \mathrm{route}(e;B)).$$

The present element reference ledger does not instantiate it. The separately declared finite model in §6 executes its own documented predicates and histories; it is not an element-wide D.E.C. ledger. Native execution would require bound Q_4 Hamming-1 traversals, scope and window, admissibility and exact weights, normalized kernels on nonempty positive-weight support, and reversed-edge SADAR pairs:

$$\sum_e P(e;B) = 1, \qquad \mathrm{adm}(e;B) = 0 \Rightarrow P(e;B) = 0, \qquad \mathfrak{S}_B(C_{\bar{e}}) = -\mathfrak{S}_B(C_e).$$

An empty admissible support does not define a normalized kernel. Each retained-return witness must bind its own retained identity, scope and lineage. Fold and comparison consumers must additionally preserve explicit compatible maps between their distinct scopes and lineages. None of these obligations is discharged by this reference's assignment tables.

Falsification. Actual negative probes must reject changed values, duplicate or missing identities, mismatched mirrors, invalid status promotion and altered provenance without replacing the input bytes. In particular, a forged certificate-like string must not turn pending admission into a pass. Unknown command flags must fail, and the verification command must not regenerate corrupted input.

Publication checks bind Main source/PDF/AUX/FLS before resolving semantic references. Figure checks cover all 126 marks at final size, including the eight unfilled slots. These are distinct from native dynamics or empirical tests.

Provenance. Main §5.17.1, Main §5.17.4, Main §5.17.5, Main §5.17.7, Main §5.17.8, Main §5.17.10, Main (66), Main (96).

Literature note. D.E.C. row grammar and execution protocol: released AOD Manual v41.02, §4 [19].

6 Folded Construction and Relational Time

Record class and data. The two-face example is an exact geometric calculation. The recurrence and response examples are declared model records: their inputs are oriented paths, retained starts, scope maps, update cuts and contact/capacity rules; their outputs are traversal counts, rational cadence, lag and signed response. No measured target enters these calculations. The exact comparator and provenance belong to each stated example.

6.1 Distinction, handed hinge and retained lineage

Setup. The schematic sameness presentation $\cdots xxx \cdots = x$ belongs to Main §1.1. Its occurrences are not previously distinguished events or ticks. The first distinction and its anchor precede the finite construction below. On an oriented construction, the shorthand $x(x)x$ resolves as $x^{-}(x^{0})x^{+}$: branch, retained hinge and counterbranch. Reflection exchanges the sides and preserves the hinge. This Manual instantiates those relations; it does not replace Main's ontology with a runtime string-reduction rule.

Derivation. Handedness records branch order through the hinge. The support accessor inherited from Main (31) is

$$RD_B(C) = \Pi_C^{-} + H_{\mu,C} + \Pi_C^{+}.$$

Equal scalar support does not identify oppositely oriented constructions. A duration realization, a count of traversals and a count of retained returns require their respective maps. In particular, a structural hinge is mandatory even when optional hinge dwell is zero; it need not be a seventh Hamming-1 traversal.

Folding relates distinct inner and outer scopes through explicit maps

$$f_{-} : (B_{-}, i_{-}, \Gamma_{-}) \longrightarrow (B_o, i_o), \qquad f_{+} : (B_{+}, i_{+}, \Gamma_{+}) \longrightarrow (B_o, i_o).$$

These maps preserve the source identity, scope, lineage, orientation, history and certificate. A proposed outer boundary requires its own retained start and return witness Γ_o. Exposure follows the maps back to the inner records. Neither common endpoint coordinates nor a handedness flip supplies an outer certificate. Cross-scale scope equality is not required.

Provenance. The source obligations are Main (5), Main (6), Main (46), Main (41) and Main (42).

6.2 An injective two-face example

Setup. Declare six distinct vertices of Q_4, with each committed base traversal changing one coordinate:

vertex	A	B	C	D	E	F
address	0000	1000	1100	0100	0010	1010

Take the oriented square faces $C_- = (A, B, C, D, A)$ and $C_+ = (B, A, E, F, B)$. They share only the coordinate edge AB, with opposite orientation. Assign interface roles to A, B, negative-side roles to C, D, and positive-side roles to E, F. The role projection is separate from the coordinate path.

Derivation. Cancellation in the oriented boundary chain gives

$$\partial(C_- + C_+) = (A, E, F, B, C, D, A).$$

The shared interface remains present in the incidence and history records. Cancellation of its outer coefficient does not erase that record. The two legs (A, E, F, B) and (B, C, D, A) each contain three Hamming-1 traversals. For n declared repetitions, let x denote the number of committed traversals in one leg family. Then

$$x = 3n, \qquad 2x = 6n. \tag{19}$$

Here n counts repetitions of this specified pattern; one counted tick is one committed Hamming-1 traversal. The factor three follows from the enumerated edges, not from naming three roles. The persisted illustrative ledger contains three repetitions, nine traversals per leg and eighteen distinct event identities. It asserts no native closure or beat count.

Specialization. Main's two-nonempty-leg bound in Main (28) constrains an already admitted closure. It neither selects this six-edge construction nor fixes a universal duration. A four-edge square is also a closed Hamming-1 walk. Conversely, closure of a geometric walk alone does not admit a native motif. One accepted retained return contributes one closure phase; later distinct returns of that same lineage supply beats as specified by Main (21). Replaying a record supplies neither a new traversal nor a new return.

If an additional model identifies each branch support operand with $3n$, its RD expression is $6n + H_\mu$, with $6n+1$ only in the single-hinge specialization. That support statement is distinct from the six-edge traversal count.

Falsification. The checker rejects the coordinate aliases $E = D$, $F = C$ before counting: they make the face boundaries coincide with opposite orientation rather than leave the claimed six-edge boundary. It also rejects a reversed interface, a non-Hamming-1 edge, a missing role, duplicate external event identities, reserialized/recounted events, wrong scope or lineage, and unsupported status promotion. Checks run against persisted inputs without changing their bytes. The generator is a separate command.

Provenance. `manual2/data/m1/folding/two_square_fold.json`, `event_ledger.json` and `folding_report.json`; Main (8), Main (74) and Main (15).

6.3 Declared recurrence and mapped comparison

Setup. The separate finite model commits the next oriented edge at each update cut. A distinct start sample follows a completed return; optional hinge dwell contributes a zero-edge sample. These update cuts are the declared model's ordering domain, not seconds or a newly derived universal clock. Two inner square lineages, each with four retained returns, are embedded into each of two oppositely oriented outer lineages. The outer traces follow the six-edge boundary with their own starts and returns. Their inner maps remain explicit.

The model declares zero optional dwell for the positive outer orientation and one sample for the negative orientation. It declares phase biases $b_+ = 1/4$, $b_- = -1/4$, four comparison contacts, and three reporting

cuts. These are fixture inputs. No target frequency, desired physical period or stochastic kernel is read by the generator.

Derivation. The inner completion cuts are $4, 9, 14, 19$. The positive outer completion cuts are $26, 33, 40, 47, 54$; the negative outer cuts are $27, 35, 43, 51, 59$. Thus both outer traces have six Hamming-1 traversals and the same declared support sum $6 + 1 = 7$, while their completion intervals are respectively seven and eight update cuts. Support and recurrence intervals retain different types even where their numbers coincide.

Each return witness binds its own start, hinge, events, scope and lineage. Event and return identities are unique; the first retained return supplies the origin, and subsequent returns supply beats. Two identity-preserving maps bind committed return prefixes to the comparator scope. Repeated contacts are required. For the windows in Table 5, define

$$\mathrm{biz}_{+|-} = \frac{\Delta \,\mathrm{bip}_+}{\Delta \,\mathrm{bip}_-}, \qquad \psi = b_+ - b_- = \frac{1}{2}, \qquad \Lambda(v) = N_+(v) - N_-(v) + \psi.$$

Here $N_\pm(v)$ are cumulative reclosure counts from each lineage's origin; they equal the window beat increments for these windows starting before both outer origins. A later window does not reset unwrapped lag. The reference counts are positive. Exchanging the comparator direction gives reciprocal cadence and negated bias/lag. Equal cadence in the first window does not remove its nonzero bias or lag.

Table 5: **Executed declared-model comparisons.** Drift is the change from the preceding reported unwrapped lag; its first value is undefined.

Window	Beat counts $(+, -)$	Cadence ratio	Unwrapped lag	Drift
$(20, 43]$	$(2, 2)$	1	$1/2$	undefined
$(20, 47]$	$(3, 2)$	$3/2$	$3/2$	1
$(20, 54]$	$(4, 3)$	$4/3$	$3/2$	0

Specialization. The declared contact support is two traversals plus one structural hinge, giving three support units. An explicit additive contact-duration map adds reflection counts $(1, 1)$ in one direction and $(2, 2)$ in the other. Its directional duration counts are five and seven. These are contact durations under a declared map, not the outer support accessor or empirically measured periods. The model executes finite Hamming-1, hinge, own-return and fold-map predicates. A native atomic realization additionally requires its complete motif and physical state construction.

Provenance. `temporal_model.json` declares the update and comparison inputs; `temporal_trace.json` stores events, witnesses, maps and contacts. Source anchors: Main (54), Main (55), Main (58), Main (61) and Main (63).

6.4 Causal returned-current response and exposure

Setup. For the same committed comparison histories, declare boundary weight $C_3 = 1$, oriented current $i = 1$ and forward duration participation $\rho^D_\omega = 5$. Use the preceding signed lag in the inherited returned-current expression. The three compatible capacities are declared inputs $10, 10, 3$, not predicted material properties. Responses commit strictly after the reporting cut that supplied their inputs.

Derivation. This specialization evaluates

$$\mathfrak{S} = C_3 \rho_\omega^D \Lambda i, \qquad P^{\mathfrak{S}} = C_3 \rho_\omega^D |\Lambda|, \qquad X = \max(0, P^{\mathfrak{S}} - C^{\mathrm{close}}).$$

It yields $\mathfrak{S} = 5/2, 15/2, 15/2$, residuals $-15/2, -5/2, 9/2$, and excesses $0, 0, 9/2$. The first two windows have nonzero lag and no positive excess. A separately bound reversal pair preserves boundary, window, history and duration participation while reversing edge orientation and signed current; its returned-current values are antisymmetric. This pairing is distinct from exchanging the two directional duration maps.

The responses commit at cuts $44, 48, 55$. The last loses positive-outer retention under the declared capacity rule, records an excess route and exposes the preserved inner maps and certificates. This is a model bridge between a typed excess and loss of outer retention; it is not an identity valid for every boundary. Identifying routed-material reclosure or radiation requires a further construction and observation map.

The constructed-field example in §9.5 retains constituent formation witnesses and evolving aggregate contacts; equal final count ratios can conceal different return timings. The separate §9.6 uses committed responses to change later retention while preserving the exposed inner wave.

Falsification. Persisted tests reject forged finite certificates, missing structural hinge, omitted/substituted inner maps, recycled inner returns, replayed or repackaged returns, one-contact comparisons, forced duration symmetry, same-step self-causation, mismatched reversed-edge witnesses and unsupported status changes. The finite model's exact reproduction is tested independently of native admission. A source-derived dynamics rule, complete motif/accessor admission and an atomic observation map are still required before these fixture cadences or capacities can become physical predictions.

Provenance. `temporal_report.json` separates declared execution, finite witness scope, native admission and empirical status. Source anchors: Main (94), Main (96), Main (101), Main (107) and Main (108).

Literature note. The inherited distinction, fold and recurrence definitions are in the released AOD foundation [19]. The finite fixture and update model in this section are Manual applications.

7 Hyperfine Target Records and Observation Map

Record class and data. The atomic inventory supplies measured target records and a calibration anchor. Its distinctions are isotope, charge, electronic and coupled states, transition, acquisition coordinate, source and uncertainty. The reported H/Cs ratio is a derived external value. The native model-to-observable map is a separate construction; an O0 comparison uses it only with a compatible measured target.

7.1 Atomic identity and sourced recurrence

Setup. The proposed application compares recurrent atomic and oscillator systems through one common fusion/reclosure model. Element placement alone does not identify a hyperfine state: hold Z, isotope A, charge, electronic state and nuclear spin fixed, and distinguish the coupled states. The retained H/Cs inventory has two neutral isotopes, four states and two transitions. The supplementary D/T target records below have separate source and reference bindings. Neither inventory adds a placement to the 118-element reference or changes a reference-A input.

Table 6: **External atomic inventory.** Frequencies are source records, not predictions of the finite folding model. Both clock transitions have $m_F = 0$ at each endpoint.

Nuclide	Electronic state; I, J	Transition	Frequency record
^{1}H, $Z = 1$	$1s\,^2S_{1/2}$; $1/2, 1/2$	$F = 1 \leftrightarrow 0$	1 420 405 751.768 Hz; reported one-sigma uncertainty 0.002 Hz; historical 1970 measurement
^{133}Cs, $Z = 55$	[Xe]$6s\,^2S_{1/2}$; $7/2, 1/2$	$F = 4 \leftrightarrow 3$	9 192 631 770 Hz; exact defining value; realization uncertainty is a separate quantity

Derivation. Hellwig et al.'s printed page 200 reports the hydrogen value with uncertainty 2×10^{-3} Hz. The exponent is bound to the retained page image, correcting the earlier plan's 0.02 Hz transcription. This is a dated measurement, not a current recommended value. Dividing the quoted decimal by the defining caesium value gives the external reference ratio

$$r^{\mathrm{ref}}_{\mathrm{H/Cs}} = 0.1545156803086000, \qquad u_r^{\mathrm{reported}} \simeq 2.1757 \times 10^{-13}.$$

The implementation retains exact rational arithmetic of the transcribed decimals; the measured hydrogen frequency is not physically exact. The uncertainty shown divides the paper's reported one-sigma uncertainty by the exact defining denominator. It is not a new uncertainty evaluation or an independent recombination of the two hydrogen evaluations.

The inventory retains the common NBS-III caesium reference and the paper's wall, cavity-pulling, spin-exchange, magnetic-field and second-order Doppler correction context. Quantitative covariance and separate realization uncertainties remain unresolved where not supplied. A missing quantity is recorded as unknown, not zero. A new precision comparison must supply the applicable conditions and shared-reference uncertainty budget.

Specialization. Atomic interrogation prepares coherence and compares its later phase with an oscillator; population measurements provide feedback to that oscillator. The counted recurrence is the stabilized oscillator cycle. A detection count, Rabi oscillation, spontaneous waiting time and hyperfine frequency are different observables. Relating oscillator cycles and quantum phase to certified native reclosures requires a physical observation map.

Provenance. `manual2/data/m1/hyperfine/target_inventory.json` binds nuclide, state, transition, source and uncertainty records. `sources/hellwig1970-page200.png` preserves the transcription evidence. The report retains the ratio's measured-input scope.

Literature note. Historical measurement and state labels: Hellwig et al. [5]. Atomic data: NIST [14, 20]. Defining reference: BIPM [1]. Clock operation: NIST [12].

7.2 Hydrogen-referenced deuterium and tritium data

Setup. These are measured target records for the ground-state hyperfine separation, with derived-external hertz coordinates. They are not D0 fixtures or evaluated O0 residuals. The underlying experiments compare D/H or T/H maser frequencies; their hertz values inherit a hydrogen reference. A later rescaling of a tritium result is the same experiment on a changed reference convention, not another independent measurement.

Table 7: **Supplementary hyperfine target records.** Parenthesized uncertainties retain the source's last-digit notation. Kramida interprets the evaluated T uncertainty as one standard deviation, with a caveat about recovering older authors' conventions. The D public abstract supplies no corresponding qualification. Distribution, coverage probability and joint covariance remain unspecified.

Isotope	Reported frequency / Hz	Reference and source	Record distinction
^{2}H	327 384 352.5222(17)	Wineland–Ramsey (1972), with H stipulated as 1 420 405 751.768 Hz	Primary public-abstract value; reported uncertainty digits give 0.0017 Hz
^{3}H	1 516 701 470.775(7)	Kramida (2010), rescaling Mathur et al. (1967) to the adopted H reference	Evaluated external value; evaluator standard uncertainty 0.007 Hz

The deuterium paper's exact-reference assumption is conditional to its reduction; it does not set measured hydrogen uncertainty to zero. Kramida's adopted H value carries parenthesized uncertainty 0.001 Hz, a distinct record from Hellwig's reported 0.002 Hz one-sigma uncertainty. Shared reference contributions and covariance must remain attached when these targets are compared. No source uncertainty is added a second time.

The deuterium public abstract is retained as downloaded publisher HTML. The tritium value above is transcribed from the NIST-hosted evaluation; the stored extraction identifies its printed pages and does not authenticate an undownloaded original PDF. The original tritium numerical result and its original H reference remain untranscribed. Experimental magnetic sublevels, preparation and acquisition windows are not supplied by these records. In particular, the H/Cs table's $m_F = 0$ assignment does not extend to deuterium.

Provenance. `manual2/data/m1/external_transitions/catalog.json` separates primary, evaluated and untranscribed records, local source hashes, reference dependence and display scope. The retained full Hellwig PDF matches the original source hash. These source records do not enter the D.E.C. calculation.

Literature note. Wineland and Ramsey [24]; Mathur et al. [11]; Kramida [9].

7.3 A common model and its atomic observation map

Setup. The application hypothesis is a reconfiguration between two states of the same isotope, bound to one frozen fusion/reclosure model. Proposed before/after configurations must preserve the isotope identity and the actual retained inner lineages. A transition need not increase element number, mass number or folding depth. A detected transfer can be identified as native shedding only after its source, destination, direction, window and measured observable are mapped.

Derivation. The atomic inventory (P4a), applicable construction (P2) and coupled recurrence (P3) are all prerequisites of the physical observation map (P4):

$$(\mathrm{P4a}, \mathrm{P2}, \mathrm{P3}) \longrightarrow \mathrm{P4}.$$

The finite declared-model checks of §6 instantiate selected construction and comparator predicates. They do not instantiate P4: its operands include the isotope-specific admitted configuration, physical state correspondence and coupled update law. Chirality or a ternary role cannot be equated with m_F

by visual resemblance, and a separate fitted law for each spectral line cannot count as a prediction of one common model.

The next native AOD physical comparison requires admitted atomic configurations, their recurrent histories and coupling records, a map from measured phase/cycles to native phase/returns, a metrological duration or frequency map, and its calibration and uncertainty model. Until those premises are established, the native predicted frequency ratio and empirical score remain absent. Separately declared conditional models and conventional benchmarks can execute under their own identified premises, as in §8. Observer phase is recorded in turns; an angular display uses $\theta = \tau\phi$, with τ the full-turn angular constant.

The constructed cadence and witnessed retention of §9.5 and §9.6 supply separate count-domain examples. A parent-loss witness in the retention model does not identify a hyperfine transition or an experimental decay endpoint.

Specialization. The hydrogen target was exposed during the discussion and supplied plans before this implementation. `target_exposure.json` records exposure by 26 October 2026; an earlier exact first-exposure time is unknown. The known pilot is retrospective. A subsequent model freeze does not establish independent selection or blindness. A calibrant cannot also be scored as a held-out target. Independent prediction requires a separately justified target and selection history. Frequency-only data do not test phase bias, lag or drift.

Falsification. Persisted hostile tests reject changed atomic foreign keys, the incorrect hydrogen uncertainty, omitted correction lists and covariance statuses, unsupported native or empirical promotion, a policy allowing target access during candidate selection, a calibrant labelled held out, detection/cycle substitution, and per-line fitted models advertised as independent predictions. These are actual schema and provenance guard tests. Their success is not empirical agreement. Every consumed status is checked; missing statuses fail rather than silently becoming zero or success.

Provenance. `observation_map.json` records the P4a/P2/P3 dependencies and pending physical map. `hyperfine_report.json` lists unresolved dependencies and null prediction/score fields. Native recurrence, comparator and duration definitions are inherited from Main (61), Main (63) and Main (57).

Literature note. External atomic observations remain downstream of the released AOD definitions [19]; the historical target remains bound to Hellwig et al. [5].

8 State-Dependent Evolution and SI Report Maps

Record classes and data. The state-dependent histories are declared model records with exact event and count coordinates. The synthetic SI examples are metrology records with explicitly stipulated observation maps and joint uncertainty. Source reductions retain measured-input provenance. The O0 diagnostic comparisons bind a declared countermodel map and external target; they do not identify that countermodel as an atomic law. Each subsection supplies its own input, report coordinate and comparator.

8.1 A response that changes later events

Setup. The second finite model retains the two-face geometry of §6. Its state contains the oriented boundary position, structural-hinge sample, retained outer identity, committed return and contact records, pending response and inner scope maps. The declared update consumes that state at the next integer cut. It instantiates the form of Main (54); the particular sequencer and capacity rule are application inputs.

Derivation. At a comparison cut, the engine calculates a burden from the committed histories and records a response for the following cut. A retention loss then removes the affected outer identity from subsequent traversal and contact eligibility. An unfinished traversal is not promoted to a return. Previously completed returns and the inner maps remain in the trace. Re-admission would require a separate construction; this model supplies none.

The controlled pair changes only the declared capacity. Its common prefix is identical before the response, while later return and contact records differ. This establishes causal state dependence within the finite model. The frozen first version remains available as the original response-calculation fixture; its precomputed histories are unchanged.

In the persisted pair, capacity changes from 100 to 2 at cut 40, and loss commits at cut 41. The affected returns at cuts 47, 54, 61 and 68 disappear; completed contacts decrease from eight to two. With the first completion retained as the lineage anchor, the finite-window reclosure-count ratio changes from 6/5 to 2/5. These are counts over the declared window, not a stationary frequency estimate.

The separate R03 model in §9.6 compares signed response with nonnegative burden at fixed raw capacity 5. Its square-return protocol and capacity do not transfer to the aggregate R02 parent of §9.5 without a separately declared adapter.

Specialization. Integer cuts, six-edge boundary traversals, hinge support, completed returns and cadence remain distinct types. The finite comparison uses Main (61) with its explicit histories and windows. It does not identify an outer recurrence with either an atomic transition or one SI second.

Falsification. The verifier rejects a post-loss affected return, an unavailable contact, altered earlier history, missing inner map, forged status or stale source binding. A changed future response cannot modify an earlier committed prefix. These checks operate on persisted counterexamples and do not repair their inputs.

Provenance. `manual2/data/m1/stateful/`; the separate model and trace verifier preserve `manual2/data/m1/folding/`. Source ownership follows Main (52), Main (53) and Main (54).

8.2 The atomic observable is an additional construction

Setup. The designated experiment uses Cs-133 as the frequency-scale reference and H-1 as the comparison target. Both configurations must belong to one declared atomic dynamics and observation law. The law must calculate a dimensionless ratio $\widehat{R}_{\mathrm{AOD}}$, before the hydrogen evaluation value is read. The external state inventory of §7 identifies the transitions; it does not supply that law.

Derivation. Once the physical observation map is established, ideal-reference reporting gives

$$\widehat{\nu}_{\mathrm{H}} = \widehat{R}_{\mathrm{AOD}}\,\Delta\nu_{\mathrm{Cs}}, \qquad \Delta\nu_{\mathrm{Cs}} = 9\,192\,631\,770\ \mathrm{Hz}, \qquad \widehat{T}_{\mathrm{H}} = \widehat{\nu}_{\mathrm{H}}^{-1}. \tag{20}$$

The Cs definition fixes the unit scale. It does not determine $\widehat{R}_{\mathrm{AOD}}$. Likewise, a ratio of finite return counts becomes a physical frequency ratio only through the declared atomic observation map.

Specialization. The cited source premises and the present atomic contract do not supply an instantiated Cs/H configuration-to-observable law. The executable underdetermination experiment therefore keeps its finite histories and Cs anchor fixed while varying the unconstrained observation relation. Different resulting hydrogen reports demonstrate that these premises do not select a unique atomic prediction. The alternatives are diagnostic countermodels, not proposed measured frequencies or fitted atomic laws.

Specifically, let r be the finite count ratio and assign the two lineages hypothetical labels $Z = 1, 55$. The unselected observation family

$$g_\lambda(Z) = 1 + \lambda\frac{55 - Z}{54}, \qquad \lambda > -1, \qquad R_\lambda = r\frac{g_\lambda(1)}{g_\lambda(55)} = r(1 + \lambda) \tag{21}$$

preserves the Cs normalization for every member while allowing every positive ratio. The witnesses $\lambda = 0, 1$ give $6/5, 12/5$ for the baseline and $2/5, 4/5$ after intervention. This proves underdetermination by the implemented finite constraints, not a general impossibility of an atomic AOD construction.

The minimum additional premise is a justified common atomic construction and observable law that fixes the relative Cs/H response without importing the tested hydrogen splitting, its reciprocal period or an equivalent gain. Selecting one countermodel by agreement with the known target would fit that target. The finite count constraints above do not select that common atomic law.

Calibration. For the conventional benchmark, dimensionless spin operators obey

$$\frac{H}{h} = \frac{A_{\mathrm{hfs}}}{h}\,\boldsymbol{I}\cdot\boldsymbol{J} + \frac{\mu_B}{h}B(g_J J_z + g_I I_z), \qquad \nu(B) = \sqrt{\Delta^2 + \left[(g_J - g_I)\mu_B B/h\right]^2} \tag{22}$$

in the $J = 1/2, m_F = 0$ block, with $A_{\mathrm{hfs}}/h = \Delta/(I + 1/2)$. The imported zero-field splittings are $\Delta_{\mathrm{H}} = 1\,420\,405\,751.768\,\mathrm{Hz}$ and $\Delta_{\mathrm{Cs}} = 9\,192\,631\,770\,\mathrm{Hz}$. The declared approximation uses $g_J = 2, g_I = 0$, omitting nuclear and bound-state corrections. At the stipulated field $B = 10^{-4}\,\mathrm{T}$, the computed shifts are approximately $2758.292\,\mathrm{Hz}$ for H and $426.200\,\mathrm{Hz}$ for Cs. These are conventional numerical benchmark outputs, with incomplete field and approximation uncertainty.

The separately labelled conventional hyperfine calculation uses this spin Hamiltonian and imported transition constants to check numerical machinery. Reproducing an imported splitting verifies that calculation; it does not predict the splitting from AOD. A conditional ideal-reference result and an actual clock realization have different uncertainty inputs. The reference definition carries no uncertainty; an employed device, transfer or calibrated observation map does.

Falsification. The prediction stage binds model, histories, observation assumptions and calibration hashes before the comparison stage reads the target. Changing an evaluation value must leave the prediction artifact unchanged and alter only the corresponding residual. A controlled finite-dynamics intervention must affect the diagnostic mapped observable at a later cut. These software properties do not supply the missing atomic law. The known hydrogen target remains retrospective; frequency and reciprocal period are two reports of the same comparison.

Provenance. `manual2/data/m1/atomic/` stores separate calculation and comparison records and the construction obligations. The frozen inventory remains under `manual2/data/m1/hyperfine/`.

Literature note. Defining scale: BIPM [1]. Historical target: Hellwig et al. [5]. Hyperfine Hamiltonian: Steck [21]. Magnetic conversion: NIST [13].

8.3 Elapsed seconds and a joint uncertainty budget

Setup. For a committed reference window ω, let C_r be the physical cycle accumulation established by an observation map, f_0 the fixed nominal reference frequency, $\bar{y}$ its effective fractional offset over that window, and d the signed correction from observed to intended endpoints. Native reclosure count and physical oscillator count are distinct inputs. All shared calibration parameters retain their covariance.

Derivation. The declared measurement equation and its first-order sensitivities are

$$\widehat{T}(\omega) = \frac{C_r}{f_0(1+\bar{y})} + d, \tag{23}$$

$$J = \left(\frac{1}{f_0(1+\bar{y})}, -\frac{C_r}{f_0(1+\bar{y})^2}, 1 \right), \qquad u_c^2(T) = J\Sigma J^{\mathsf{T}}. \tag{24}$$

The elapsed interval $\widehat{T}(\omega)$ is distinct from the transition period $\widehat{T}_{\mathrm{H}}$ in (20). A faster reference gives a shorter inferred interval at fixed accumulated cycles. An ideal frequency divider transforms counts, nominal frequency and covariance together and preserves the reported interval and its uncertainty.

Specialization. A shared endpoint offset cancels only when the measurement equation takes its difference. A common frequency bias need not average down. A quadratic field shift at zero mean field can have nonzero expectation and uncertainty even where its first derivative vanishes. Unresolved discrete cycle slips require an ambiguity model. Unknown covariance, drift or transfer effects are not numerical zeros.

Calibration. Each budget component records its unit, source, validity window, correction sign, covariance group and inclusion in any aggregate. An already included realization contribution is not added again as an independent term. The reported measurand distinguishes local proper time from a coordinated time scale; only applicable external motion, potential and transfer corrections enter. A later calibration may revise a retrospective report, but it cannot be represented as available to an earlier online computation.

The frozen v2 records retain first-order diagnostic budgets and their original one-percent marginal guards. Those guards do not prove nonlinear adequacy: the M3 successor checks exhibit a near-cancelled product whose exact variance is 50.90005 times its first-order value, and a zero-derivative interval with nonzero exact variance. Main (165) supplies full finite-law propagation when that joint law is supplied; covariance alone does not complete a nonlinear model.

The persisted positive examples exercise declared synthetic maps and uncertainty budgets. They do not invent a clock calibration certificate. A materially incomplete physical budget produces a scoped partial result; independent finite-model execution continues.

One exact synthetic budget yields $\widehat{T} = 1\,\mathrm{s}$ and $u_c^2 = 4904\,\mathrm{ps}^2$, hence $u_c \simeq 70.0286\,\mathrm{ps}$. A separate integration example maps five reference reclosures to one second by stipulation; the controlled intervention changes the compared count from six to two while preserving those five reference reclosures. Neither stipulation supplies a measured atomic observation gain.

Falsification. The integration checks reject covariance with incorrect units, ordering, asymmetry or a negative quadratic direction; unknown material terms, duplicate aggregate inclusion, stale scope and future calibration are tested separately. Nonlinear and discrete cases cannot acquire a complete precision claim by using only a linear derivative.

Provenance. `manual2/data/m1/metrology/`; the application contract is preserved in `manual2/evidence/v2-request/SI_ANCHOR_CONTRACT.md`.

Literature note. Realization and proper time: CCTF [2]. Measurement models and correlated uncertainty: JCGM [6, 7].

8.4 Source-bound metrological comparisons

Setup. The M3 application binds the R03 observations of §9.6 and the four diagnostic outputs of (21). Each calculation retains its original inputs before the historical hydrogen target is read. The target's measurement interval has no established correspondence to the simulated cut window.

Derivation. The inherited hydrogen reduction evaluates Hellwig et al.'s Tables IV and VI with their stated correction ownership. Their corrected values are 1 420 405 751.7691 Hz and 1 420 405 751.7667 Hz. Their exact printed-input mean, 1 420 405 751.7679 Hz, remains distinct from the published rounded mean and its reported 0.002 Hz uncertainty. A correction already included in an input is not added again.

The comparison subtracts the authenticated target from each diagnostic output. It retains the reference dependencies, correction ownership and unresolved applicability and uncertainty. It selects neither gain witness and infers no apparatus correction from a residual.

Specialization. The D/T records in §7.2 supply hydrogen-referenced external frequencies. An atomic comparison still requires the preparation, configuration-to-observable law and matching apparatus inputs. These acquired targets do not alter the frozen application profile or supply a native isotope-specific frequency. The selected Lv-290 intervals in §9.2 instead concern detector events.

Calibration. The Cs defining frequency supplies the scale. The NIST-F2 August 2013 uncertainties and typical Table 1 budget retain separate campaign identities; their printed fractional values are converted to hertz using that scale. Shared reference contributions retain their dependence. Complete physical uncertainty and the map from reflection duration to the measurement interval remain unspecified.

Falsification. Changing an evaluation target must preserve the frozen prediction. Substituted histories, duplicate corrections and unsupported promotion of unknown terms invalidate the corresponding record. Exact endpoint subtraction alone does not establish a matched physical observation.

Provenance. `manual2/data/m1/m3_metrology/` separates predictions, comparisons and the application report. Typed mappings use Main §7.3; exact propagation requires the joint law of Main (165).

Literature note. Hydrogen source reduction: Hellwig et al. [5]. Fountain realization: Heavner et al. [4].

8.5 Published frequency covariance

Setup. The BIPM adjustment input retains 153 rows, including seven effectively excluded by enlarged uncertainties, and 1,168 correlation coefficients. The output contains 14 frequency estimates and their covariance. Raw bytes, source metadata and prior target exposure are preserved.

Derivation. The source README describes a ratio basis, while the numerical matrix is consistent with squared hertz. Under this documented interpretation, ratios to the exact Cs definition use $\Sigma_{\nu/\nu_{\mathrm{Cs}}} = \Sigma_{\nu}/\nu_{\mathrm{Cs}}^2$; general ratios use their full Jacobian. Exact rational intervals propagate the nearest-rounding cells of covariance inputs and printed frequencies. All 105 calculated uncertainty intervals overlap the corresponding squared output-uncertainty rounding cells. This includes cases where a point estimate alone lies outside the last-digit interval of the output. The preserved check used a nearest-rounding cell for every token. The separate M3 precision successor identifies 26 printed-zero cells affecting 13 ratios: their formatting or underflow convention is not supplied by the source. Treating those zeros as point transcriptions is a conditional numerical check, not a source-exact uncertainty claim. All 105 point results are preserved; a one-percent sigma mutation in the selected affected ratio is rejected under that conditional policy.

Specialization. The recommendation multiplies standard uncertainty by 2.35, hence covariance by 2.35^2. This is distinct from a coverage factor. The adjustment's added fractional term $5.324065020 \times 10^{-17}$ is already represented in its outputs and is not appended again. Adjusted estimates and their contributing measurements are not independent observations of one another.

Falsification. Hash checks preserve the input; independent parser and numerical tests reject altered identities, correlation domains, covariance, units and output uncertainties. The numerical unit interpretation is not an official corrigendum, a reconstruction of the least-squares adjustment or a physical clock realization.

Provenance. `manual2/data/m1/bipm/source_manifest.json` , `bipm_report.json` and the separate `bipm_precision_report.json`; 153 source records and all 105 uncertainty comparisons remain inspectable.

Literature note. Published input, output, covariance and analysis notes: BIPM [10].

9 Isotope Half-Lives and a Common Survival Experiment

Record classes and data. Finite evaluated half-lives and individual detector intervals are measured target records, with state, quantity, source and uncertainty preserved. Stability qualifications, unknown values, unobserved declarations and rejected candidates retain their distinct inventory records. Nucleon grouping is exact bookkeeping. The survival, constructed-coupling and retained-return applications are separately declared models, each with its own history, coupling, capacity and report coordinate. An O0 physical comparison requires a common observation map; a half-life table alone does not supply the model's state-dependent loss rule.

9.1 Source records and nucleon bookkeeping

Setup. The element reference remains the 118 placements and 126 slots of §3. A separate isotope layer joins by atomic number and symbol. It retains the nuclear-state assignment, selected evaluation, original notation, asymmetric uncertainty, decay channels and experimental overlap. Total half-life, channel-partial half-life and an individual waiting time are distinct quantities. Observational stability supplies no measured infinity.

Derivation. For the formal grouping of a record with proton count Z, neutron count $N = A - Z$, define

$$q = \min(\lfloor Z/2 \rfloor, \lfloor N/2 \rfloor), \quad r_p = Z - 2q, \quad r_n = N - 2q, \qquad A = 4q + r_p + r_n. \tag{25}$$

These are exact integer coordinates. Identifying a group with a physical alpha cluster requires an additional construction. The bookkeeping alone supplies neither a loss probability nor a half-life.

Specialization. Tables 8–10 keep the printed reference compact. The companion retains the broader family records, source-specific updates and uncertainty qualifications. H-1 and Cs-133 retain their recurrence roles. Fe-56 is the selected binding benchmark; the Ni-62 digital control prevents calling Fe-56 the maximum binding energy per nucleon. Pb-208 remains observationally stable, with its experimental alpha-channel limit stored separately from total lifetime.

Isotope	Z, N	q	r_p, r_n	Adopted $T_{1/2}$
^{1}H	$1, 0$	0	$1, 0$	observationally stable
^{133}Cs	$55, 78$	27	$1, 24$	observationally stable

Table 8: Recurrence anchors. Stability is an observational classification; no infinite half-life is assigned.

Isotope	Z, N	q	r_p, r_n	Adopted $T_{1/2}$
^{56}Fe	$26, 30$	13	$0, 4$	observationally stable
^{208}Pb	$82, 126$	41	$0, 44$	observationally stable

Table 9: Binding and observational-stability controls. The digital reference retains Ni-62 and the channel-specific Pb-208 bound.

Isotope	Z, N	q	r_p, r_n	Adopted $T_{1/2}$
^{294}Og	$118, 176$	59	$0, 58$	$0.58^{+0.44}_{-0.18}$ ms
^{290}Lv	$116, 174$	58	$0, 58$	$8.3^{+3.5}_{-1.9}$ ms
^{286}Fl	$114, 172$	57	$0, 58$	166^{+40}_{-27} ms
^{293}Lv	$116, 177$	58	$0, 61$	95^{+63}_{-27} ms
^{289}Fl	$114, 175$	57	$0, 61$	$2.4^{+0.8}_{-0.5}$ s
^{298}Fl	$114, 184$	57	$0, 70$	unobserved target

Table 10: Selected superheavy records. Asymmetric uncertainties are expanded from the adopted last-digit notation. Fl-298 is an unobserved target. Newer experimental analyses remain separate in the companion.

$$^{294}\mathrm{Og} \xrightarrow{\alpha} {}^{290}\mathrm{Lv} \xrightarrow{\alpha} {}^{286}\mathrm{Fl} \xrightarrow{\alpha} {}^{282}\mathrm{Cn} \xrightarrow{\mathrm{SF}} \text{fragments}$$
$$^{293}\mathrm{Lv} \xrightarrow{\alpha} {}^{289}\mathrm{Fl} \xrightarrow{\alpha} {}^{285}\mathrm{Cn}$$

Figure 2: Observed alpha relationships in the selected records. The first route is one branch: Fl-286 also fissions. Cn-285 has further alpha decay; it is a scope endpoint here, not an asserted terminal stable state.

Calibration. Evaluated half-lives retain their source units and explicit conversion to seconds. Newer experimental analyses remain distinct from the adopted evaluation; overlapping events are not averaged as independent data. Fl-298 has no measured half-life and receives no empirical score. The progression Fl-289, Og-294, Fl-298 is a comparison set, not a decay chain. The sourced alpha edges in Fig. 2 obey $\Delta Z = -2$, $\Delta A = -4$; spontaneous fission does not acquire an invented unique daughter.

Falsification. Source, state, quantity and status mutations must fail without rewriting inputs. The logarithmic half-life display names the selected isotope and gives limits, stability, unknown records and uncovered elements nonnumeric treatments. It does not reinterpret the structural-role colors or an arbitrary A arithmetic input as measured stability.

Provenance. `manual2/data/m1/isotopes/` binds the source bytes, selected records, updates and generated tables. The original element ledger and historical numeric records remain unchanged.

Literature note. Evaluated isotope data: ENSDF [16]; evaluation cross-check: NUBASE2020 [8]; binding control: AME2020 [23].

9.2 Selected implantation-to-decay records

Setup. The two selected Lv-290 chains in the retained Gates et al. source begin at implantation in the detector. Their first decay intervals are 5.66 ms and 17.3 ms, respectively. Their isotope assignment, transport and event-selection context remain attached to each record.

Derivation. Exact conversion of the printed intervals gives 283/50,000 s and 173/10,000 s. These individual observed delays are distinct from formation time and ensemble half-life. The reported digitizer capability of up to 125 MS/s does not specify the actual timestamp resolution or its calibration; those inputs remain missing from a complete timing comparison.

Provenance. The event records in `manual2/data/m1/m3_metrology/` bind the retained source under `manual2/data/m1/isotopes/sources/`; the adopted half-life catalog remains separate.

Literature note. The two chains and detector acquisition: Gates et al., Fig. 2 and experimental description [3].

9.3 A state-dependent finite law

Setup. This model's data card names the two square paths, complete choice histories, reference returns, kernel weights and capacity rule. The report coordinate is survival over reference-return counts, with exact finite-law probabilities; the comparator is the declared retained/loss predicate. It uses no measured half-life as a kernel or capacity input. The third application model instantiates the conditional H1 kernel of Main (66), paired orientation response of Main (96), and excess construction of Main (108).

Main does not select a nuclear state, its kernel weights or a physical loss observation for this example. Within the declared finite scope, the application fixes two retained square paths

$$\gamma_+ = (0000, 1000, 1001, 0001, 0000), \tag{26}$$

$$\gamma_- = (0000, 0100, 0101, 0001, 0000). \tag{27}$$

Their vertices have one injective Q4 encoding. At 0000, the incoming ancestor slot is 0001; the three continuations are $1000, 0100, 0010$. The third exits through 0010, then takes 1010 or 0110. These last routes carry hypothetical alpha and fission labels, each with declared conditional weight one. Their nuclear identification is not established.

A separate reference lineage repeats γ_+. Both lineages begin with two distinct retained-start witnesses. One step of the survival experiment completes a new reference return, binds the already committed candidate prefix, normalizes the kernel, executes the selected path and commits the response for the next contact. Four H1 traversals and the retained start witness the finite square return; passing this predicate does not establish native motif admission. The structural hinge is retained. No universal seconds per traversal or scheduler cut are assigned.

Derivation. Let h be a particular retained choice history, with signed committed memory R_h, common capacity $C \geq 0$, and common positive loss gain λ. The declared specialization is

$$X_h = \max(0, |R_h| - C), \qquad (w_+, w_-, w_{\text{exit}}) = (1, 1, \lambda X_h), \tag{28}$$

$$P_+(h) = P_-(h) = \frac{1}{2 + \lambda X_h}, \qquad P_{\text{exit}}(h) = \frac{\lambda X_h}{2 + \lambda X_h}, \tag{29}$$

$$R_{h\sigma} = R_h + \sigma, \qquad \sigma \in \{-1, +1\}. \tag{30}$$

The last update retains the oriented response pair $(\sigma, -\sigma)$. Its unit magnitude specializes declared pressure 2 and signed lag magnitude $1/2$; neither is an isotope-derived support accessor or a lag inferred from measured decay. This is the explicit application premise between the source response form and the weight law. Zero exit weight leaves the exit structurally admitted and unrealized.

For exact history mass μ_h, multiplication by the probabilities in (29) supplies the child masses. Absorbed histories retain their completed return IDs and inner scope map. They have no later candidate return. Distinct prefixes remain distinct even if their numerical memories agree. Summing only retained histories after reference return c gives

$$S(c) = \Pr(C_{\text{loss}} > c) = \sum_{h \text{ retained after } c} \mu_h, \qquad c_{1/2} = \min\{c : S(c) \leq \tfrac{1}{2}\}. \tag{31}$$

Thus $c_{1/2}$ is the first discrete crossing. The recorded crossing interval $(c_{1/2} - 1, c_{1/2}]$ makes no within-cycle interpolation. If no crossing occurs by the horizon, the result is a censored lower bound. A single loss event is not this ensemble median.

Specialization. With $R_\varnothing = 0, C = 1, \lambda = 1$, exact propagation gives

$$\big(S(0), \ldots, S(8)\big) = \left(1, 1, 1, \frac{5}{6}, \frac{3}{4}, \frac{3}{5}, \frac{19}{36}, \frac{703}{1680}, \frac{211}{576}\right). \tag{32}$$

The median crosses in $(6, 7]$ reference returns. Each run expands 255 retained prefixes without merging. Doubling the common loss gain gives $(4, 5]$; setting the common capacity to zero gives $(2, 3]$. Disabling new response commits at return 2 leaves $S(c) = 1$ through return 8. Its survival agrees with the baseline

through return 2 and first differs at return 3: 1 versus 5/6. This controlled change therefore affects subsequent loss rather than merely annotating a precomputed trace.

The finite law is evaluated with the exact M3 mapping implementation and Main (165). For the retained indicator at return 8, its expectation is 211/576 and its variance is (211/576)(365/576). This ensemble spread is distinct from uncertainty about the nuclear model or the experimental calibration.

Calibration. No isotope-specific hazard or capacity is fitted. Within this model, every isotope label selects the same surrogate configuration; (25) does not change its law. A physical elapsed-time report would require an observation map taking reference returns to observed oscillator cycles. For example, the hypothetical constant gain $g > 0$ would give $T = gc/\Delta\nu_{\mathrm{Cs}}$, with $\Delta\nu_{\mathrm{Cs}} = 9\,192\,631\,770\,\mathrm{Hz}$. The unselected gains $g = 1, 2$ give different times while preserving all native histories and the same SI definition. The defining frequency alone therefore cannot select a decay-time prediction. Realization, model discrepancy and covariance remain unresolved rather than numerical zeros.

Falsification. The exact tests reject probability leakage, coordinate aliases, lost lineage, a forged return, nonantisymmetric response, later survival disconnected from feedback, and status promotion. The target-free prediction runs without the isotope catalog. The comparison replays that complete prediction before consuming validated source records. A changed target or a self-consistent replacement hash cannot rewrite the frozen prediction.

Provenance. `manual2/data/m1/survival/` records the common model contract, four exact runs, prediction, comparison and unresolved construction obligations. The earlier finite folding and stateful fixtures remain separate. The observation-gain dependency instantiates Main §7.5. The model identity is `m1:finite-history-retention-survival:v3`.

Literature note. SI reference definition: BIPM [1].

9.4 A measured test of the restricted specialization

Setup. Equal surrogate states with a common observation and clock map imply identical survival distributions and a half-life ratio of one. This restricted consequence is testable even though the isotope-to-native-state construction is incomplete. The selected adopted records were exposed before this calculation; this is a retrospective comparison.

Derivation. Using numerator and denominator source uncertainty envelopes without treating them as independent Gaussian inputs yields

Half-life ratio	Adopted central ratio	Source-envelope ratio range
Og-294 / Lv-290	29/415	$[2/59, 51/320]$
Lv-290 / Fl-286	1/20	$[16/515, 59/695]$
Lv-293 / Fl-289	19/480	$[17/800, 79/950]$

All three ranges exclude one. These are descriptive enclosures constructed from the reported uncertainties, not a joint confidence statement or significance score.

Specialization. The identity-insensitive, shared-clock specialization disagrees with these reported envelopes. A successful common law must therefore distinguish the isotope configurations or their justified observation relations. The measurements do not select a replacement law. This result does not falsify the general AOD construction or establish a native nuclear prediction. The next obligations are an admitted isotope/state configuration, an independently justified state-to-loss law, the reference observation map and a complete physical uncertainty model. Fl-298 remains an unscored target.

Calibration. A separate conventional benchmark imports each adopted $T_{1/2}$ and defines $S(t) = \exp(-\lambda_{\exp} t)$, $\lambda_{\exp} = \ln 2 / T_{1/2}$. Its mean lifetime is $T_{1/2}/\ln 2$, not $T_{1/2}$. Reproducing survival $1/2$ at an imported half-life checks the conventional calculation; it does not predict that half-life. The native finite law in (29) is not assumed exponential.

Falsification. A physical isotope prediction must fix its state construction and observation law before joining the evaluation target. Independent prediction additionally requires an exposure design beyond these already known records. Source-specific revisions remain separately bound comparisons, and no unobserved lifetime can be counted as agreement.

Provenance. The separate `prediction.json` and `comparison.json` bind their complete inputs and exact outputs. Tests SURV-01–19 state their finite assertion scope; passing them does not certify nuclear admission.

9.5 Constructed-field cadence

Setup. The R02 constructed-coupling law assigns 118 distinct 3:3:6 and 176 distinct 3:4:6 constituent formation waves to an Og-294 candidate. Their scope maps bind the inner certificates to an outer construction with 15,508 outward occurrences and structural $RD = 31{,}017$. The 294 inner formation histories remain fixed while five aggregate fields evolve: parent, reference and three coupled backgrounds. This is a declared aggregate reduction; the constituent assignment does not establish a nuclear configuration. The third background is present in every case; the feedback intervention changes its influence on the parent.

Derivation. Each new outer reclosure emits a contact with its committed prefix. The common sequencer sets the next action interval to the field's structural RD plus its incoming last-two-contact RCD durations; after completion, the next start follows one cut later. Forward and reverse reflection histories retain their direction, lag and contact operands. The resulting counts are

Table 11: Constructed coupling to cut 4×10^9, counted after each case's parent formation.

Case	New parent reclosures	Reference bips	Ratio
Baseline	4	1902	2/951
Added 3:4:6 feedback	4	1902	2/951
Same contact, feedback disabled	4	1902	2/951
One fewer 3:4:6 constituent	4	1909	4/1909

Added feedback changes later return timings despite the equal final ratio. The baseline ratios in successive disjoint windows are $1/513, 1/489, 1/461, 1/439$. Its horizon record has 127,571 parent H1 traversals,

including 3,507 in an unfinished return, four structural hinges and zero extra dwells. The channels record 11,212 ticks and 11,212 clicks. A label change alone leaves the trajectory unchanged.

Specialization. The baseline nonnegative load over the full window is 224,657,122. The same committed operands give signed diagnostic −224,657,122. The sum specializes Main (106); capacity remains unassigned. For that fixed load, the unselected capacities 112,328,561, 224,657,122 and 224,657,123 give respective excesses $112{,}328{,}561, 0, 0$ under Main (107) and Main (108). Neither the load nor ordinary emitted current selects a parent-loss event or a survival distribution.

Provenance. The law `matter1:constructed-coupling-r02:v1` and `manual2/data/m1/constructed_coupling/` retain the formation, scope maps, directed contacts and window records. Recurrence uses Main (54), Main (55) and Main (61).

9.6 Witnessed response, burden and retention

Setup. The separate R03 law prepares four candidate square returns, two reference returns, three background 3:4:6 returns and two outer-parent formation returns. The parent folds the evolving candidate wave; reference and background remain external. Each round completes the reference and background comparisons, decides from prior committed responses, then executes a candidate return or an open exit. An accepted return commits the response for a later decision. The raw normalization unit is $u = 5$, the reference cross-return's structural RD. Raw closure capacity is $C^{\mathrm{close}} = 5$, hence diagnostic capacity one. These choices belong to R03.

Derivation. With $C_{3,e} = 1$, active committed contributions $e \in A_h$ supply

$$R_h = \sum_{e \in A_h} \rho^D_{\omega,e} \Lambda_e i_e, \qquad B_h = \sum_{e \in A_h} \rho^D_{\omega,e} |\Lambda_e|. \tag{33}$$

The lag comes from certified return counts; orientation comes from the candidate return witness. Signed cancellation leaves the burden contributions present. The common conditional weight formula is

$$X(L) = \frac{\max(0, L-5)}{5}, \qquad (w_+, w_-, w_{\mathrm{exit}}) = (1, 1, X(L)), \qquad P_{\mathrm{exit}} = \frac{X(L)}{2 + X(L)}. \tag{34}$$

The net-memory control uses $L = |R_h|$; the burden model uses $L = B_h$. The reference contributes raw pressure 5 and lag 1. The constructed background has $RD = 68 + 1 + 68 = 137$ and baseline lag zero.

Table 12: Two-return prefixes, each with probability 1/4. Exit probabilities apply at the subsequent decision.

Prefix	R_h	B_h	R_h/u	B_h/u	Net exit	Burden exit
++	10	10	2	2	1/3	1/3
+−	0	10	0	2	0	1/3
−+	0	10	0	2	0	1/3
−−	−10	10	−2	2	1/3	1/3

Specialization. Exact path probabilities give the retained masses in Table 13. Gating feedback preserves its response history while disabling its use. Restricting active membership to the latest response group records the earlier group's exclusion; the duration-participation window remains fixed.

Neither operation erases a contribution. A relief of a fixed retained group would require its own recorded rebalancing operation.

Table 13: R03 survival after three new reference returns.

Case	$S(3)$
Net memory	5/6
Nonnegative burden	2/3
Burden input gated after response 2	1
Only latest response group participates	1
Second new 3:4:6 return held pending	5/76

Holding the background return pending after its first H1 edge changes its lag to one. The second response carries raw burden $5 + 137 = 142$; total burden 147 gives next-exit probability 71/76. This is a finite intervention under the same capacity rule. Its median loss count crosses in $(2, 3]$; the other four cases have no median crossing by the horizon.

An absorbed third-round history retains two new parent reclosures over three reference bips, two new candidate reclosures, eight candidate H1 traversals and ten outer H1 traversals. The last two outer edges form an open exit, not another return. The absorbed burden case has two candidate hinges, three outer hinges including the exit turn, zero optional dwells, six channel ticks and six clicks; the reference completes twelve H1 traversals. The loss witness exposes the complete evolving inner lineage. Surviving histories are censored at three reference returns. Individual loss count, ensemble median, current emission and physical half-life remain distinct.

Falsification. The first four cases share the committed prefix before the third decision. Reversing all current orientations reverses signed response and preserves burden. A separate reflection control gives forward duration 5 and reverse duration 7. Passive observation leaves the evolution unchanged; a future response cannot alter an earlier decision. These controls distinguish retained history, active participation and directional reflection.

The observation interface retains the complete prepared population, directed contacts, committed prefixes, pending returns and prior comparison. Its selected window projects two candidate reclosures and three reference reclosures from that history; the projection does not replace the evolving state. With the declared gain of one cycle per reclosure, the conditional frequency is 6 128 421 180 Hz. Replaying the original inputs checks the complete report. Physical cycle calibration remains a separate premise, as in §8.4.

Provenance. The law `m1:witnessed-retention:r03:integration-v1` and `manual2/data/m1/witnessed_retention/` bind the complete histories and terminal counts. The burden, residual and excess specialize Main (106), Main (107) and Main (108); their use at a later decision follows Main (101).

9.7 Two configurations under one common law

Setup. The paired construction selects Fl-286 and Fl-289 ground-state labels externally, with Og-294 as a larger control. Preparation fixes distinct two-return constituent seeds, their ordering, outer fold and common reference/background seeds. Each candidate retains its constituent witnesses, assembly maps and directed aggregate couplings. Physical preparation and internal nuclear couplings are unspecified.

Derivation. The declared 3:3:6 proton-facing and 3:4:6 neutron-facing inventories give

Table 14: Support of the three explicitly constructed aggregate candidates.

Candidate	3:3:6 waves	3:4:6 waves	Outward occurrences	RD
Fl-286	114	172	15116	30233
Fl-289	114	175	15320	30641
Og-294	118	176	15508	31017

The common R02 update and contact laws act on the five aggregate fields; the constituent formation histories remain fixed. No isotope-specific capacity, hazard or lifetime multiplier enters this construction. Counting after formation cuts 61469, 62285 and 63037, respectively, to the common cut 4×10^9 gives parent/reference counts $4/1940 = 1/485$ for Fl-286, $4/1920 = 1/480$ for Fl-289 and $4/1902 = 2/951$ for Og-294. Each parent completes four reclosures; their return cuts and reference exposures differ.

Specialization. A combined R02–R03 model requires an explicit map of states, retained scopes, event order, active contribution groups, normalization, capacity, reference exposure and the loss/exposure operation. R02's reference has structural RD 61; R03's square cross-return has RD 5. The latter's raw capacity 5 is a choice of that protocol. It does not supply a nuclear aggregate capacity. A positive burden or surplus alone supplies no loss predicate.

The adapter assessment binds each constructed state, directed contact history and reference exposure. It leaves the normalization, capacity and loss map unselected. Thus the count-domain comparison is reproducible while the combined retention law remains undetermined.

The isotope/state identification, physical update and retention law, experimental loss and elapsed-time map, and dependence-aware uncertainty remain unspecified. Fl-298 remains unscored. The previously exposed Fl measurements can support a retrospective comparison once those premises are supplied.

Falsification. Relabelling the Fl-286 construction as Og-294 leaves its identity, actions, packets and return history unchanged through cut 1,100,000. Adding three neutron-facing constituents between the paired preparations changes the parent identity and scope map under the same law. A label change and a configuration change therefore have distinct witnessed effects.

Provenance. `matter1:paired-aggregate-configurations:v1` binds the common law and separate preparations under `manual2/data/m1/constructed_coupling/`. `manual2/data/m1/retention_adapter/` records their adapter assessment; `manual2/data/m1/m3_observation/` binds the R03 observation report. The R02 and R03 examples retain their separate law identities.

10 An All-Element Inner-Recurrence Campaign

Record class and data. This is a declared model/simulation with native generic construction witnesses. Its inputs are isotope counts, constituent recipes, mapped scopes, the coupling rule and integer sampler. Its report coordinate is retained-return count in a configuration-specific window. D0 internal contact audits compare each signed current with its opposite bookkeeping view; their pair residual is zero. Measured transition frequencies and half-lives belong to separate target records and are not inputs to this dynamics.

Setup. The campaign retains the 118 element reference records and the 27 curated isotope/state records of §9. Their union contains 135 distinct pairs (Z, A). The 145 source records remain separate: an unspecified reference selection, an assigned ground state and a tentative isomer are different inputs even when their nucleon counts coincide. A preparation map must distinguish their constructed states before different dynamics can be attributed to that distinction.

For the declared neutral inventory, set $N = A - Z$ and $E = Z$. Construct distinct proton-facing 3:3, neutron-facing 3:4 and electron-facing 1:2 recurrent supports, each with two retained formation returns. Their ordered list induces a directed ring with one incoming neighbour per constituent. This adjacency is an application preparation. The ring label alone does not establish the admitted fractal field ring gate of Main §5.10.

Each inner scope maps to the proposed common outer scope. The independent outer trace and certificate then close the proposal using Main (114). The generic construction identifies retained support and lineage; identifying an electronic or nuclear preparation requires an additional state correspondence. In particular, hydrogen-1 has $N = 0$ and a valid proton/electron inventory.

Derivation. The enumerated outward occurrences of this construction are

$$B = 30Z + 68N + 4E, \qquad RD_{\text{outer}} = 2B + 1. \tag{35}$$

The final term counts the mandatory hinge. These constructed support quantities retain their own types; the static ledger's conditional support sum remains defined by its earlier model.

Let $R_i^{(r)}$ be the signed response already committed to constituent i before round $r + 1$, and let $D_i^{(r)}$ be its directional duration from the last two committed incoming contacts. The declared whole-return orientation kernel and traversal pacing are

$$w_i^{\pm} = 1 + \max(0, \pm R_i^{(r)}), \qquad P_i^{\pm} = \frac{w_i^{\pm}}{w_i^{+} + w_i^{-}}, \tag{36}$$

$$d_i^{(r+1)} = RD_i + D_i^{(r)} + \max(0, R_i^{(r)}). \tag{37}$$

The orientation selects a complete continuation recipe; its H1 traversals, hinge and return are subsequently executed and independently certified. The two-outcome recipe selection is distinct from claiming two alternatives at every atomic traversal.

A genuine inner reclosure emits a returned-current support to its neighbour. For source orientation σ_j and the receiver-minus-sender recurrence lag $\ell_{i|j}$ bound at emission, the application assigns directional reflection counts $(\max(0, \ell_{i|j}), \max(0, -\ell_{i|j}))$. The first count supplies the additional reflected extent of the directed support trace. This is the declared bridge from recurrence lag to reflection coordinates; native admission validates the resulting support rather than deriving this bridge. The last two committed supports then supply $D_{i|j}$, and the application commits

$$R_i^{(r+1)} = \sigma_j D_{i|j} \ell_{i|j}. \tag{38}$$

The opposite view carries the negative contribution. The contributing contact, its prefixes and its non-negative burden remain present when signed values cancel. Each round completes its returns and cross-returns before the next round consumes them. This realizes the committed-input order of Main §4.3.2; the pacing and orientation weights are declared finite-model choices.

After three inner rounds, the new outer certificate binds the extended inner prefixes. An independent 3:3 reference executes RD-paced traversals with a one-cut restart. Within the recorded window from the start of internal execution to the new outer return, repeated bound cross-returns supply the native cadence and lag comparison. Window endpoints depend on the particular construction. A quotient of results from different windows therefore does not by itself identify an atomic frequency ratio.

Specialization. The neutral hydrogen-1 construction has $B = 34$ and $RD_{\text{outer}} = 69$. The neutral Og-294 construction has 412 distinct constituents, $B = 15\,980$ and $RD_{\text{outer}} = 31\,961$. Neither scalar alone specifies emitted radiation or isotope survival. The finite campaign retains all simulated histories. Exact enumeration of a small orientation tree checks its conditional probability mass; larger inventories use a recorded integer sampler. A sampled trajectory is not an ensemble survival curve.

Calibration. The historical hydrogen reductions in §7 remain separate from the generic recurrence results. The SAO beat comparison consumes an emitted phase map and the apparatus window; the SI anchor supplies the scale after the model ratio has been identified. The source-specific Table IV, Table VI and combined-mean uncertainties remain attached to their respective quantities.

For isotope survival, a configuration-dependent retention/loss observation must first determine which executed event contributes to loss of the specified state. A reference-recurrence calibration then supplies physical elapsed time. The existing square-path survival fixture does not select that nuclear loss rule or transfer its capacity to the 135 inventories.

Falsification. Matched feedback controls retain the first round and alter only subsequent continuation. Omitting the observation projection preserves the executed trajectory. Original-input replay detects replaced returns, changed contacts and forged status promotion, even when local hashes are recomputed. Ground/isomer labels without a distinct preparation map cannot manufacture different dynamics. No state reduction is used; a later reduction must preserve the joint conditional law of the next reduced state and the emitted observable.

Provenance. The campaign consumes the source-bound current native admission implementation and an explicitly identified historical construction recipe. `manual2/data/m1/campaign/` contains target-free identities, original configurations, executed cases and separately produced comparisons. `manual2/evidence/campaign-source-binding.json` records the authority roles.

11 State-Resolved Element Properties

Setup. The property catalog distinguishes an element, an isotope in a nuclear state, and an atom or ion in an electronic state. Its identity fields are atomic number, isotope assignment, nuclear state, charge and electronic state. An unspecified isotope remains unspecified. A neutral ground-state ionization energy describes the transition to the singly charged continuum; the second ionization energy has a singly charged initial state. Electron affinity also requires the negative-ion endpoint. A nuclear mass and an atomic mass have different electron inventories.

Each property record binds one entity, one quantity, its unit, source locator, value qualification and uncertainty convention. The method of obtaining a value, prior exposure to the target and the scope of a model claim are separate fields. An evaluated value may incorporate a measurement or an extrapolation; neither designation establishes a native AOD prediction. A decimal point value, an interval, a one-sided limit, a text assignment and an unknown are distinct records. An unknown uncertainty is not zero.

Derivation. Let s denote a complete declared state identity, p a property and c its conditions. An empirical property comparison is admissible only after the prediction and target have been assigned the same triple:

$$(s, p, c)_{\text{prediction}} = (s, p, c)_{\text{target}}. \tag{39}$$

This is a catalog join condition, not a physical observation law. Residuals between an unselected hypothetical map and a known target remain arithmetic diagnostics; they do not pass this empirical comparison gate. In particular, an isotope-unspecified atomic evaluation cannot supply an isotope shift. A selected line requires its endpoints; a field-dependent shift also requires the field, state assignment and applicable regime. No magnetic sublevel is created from an unresolved total angular momentum label.

The constructed histories of §10 retain their formation witnesses and relational observables. Assigning an atomic state to a history and assigning an emitted phase or an energy to that state are additional maps. Distinct positive maps satisfying the declared native and dimensional constraints can assign different values to the same history. These typed witnesses show that those constraints alone do not select a physical value. A common calibration anchor supplies a scale only after the relevant dimensionless relation has been established.

Specialization. The pilot comprises hydrogen-1, hydrogen-2, hydrogen-3, helium-3, helium-4 and caesium-133. Atomic identities with unspecified isotope remain separate from these pilot assignments. The element extension covers atomic numbers 1–118. Its 500 entities carry 4 354 property records: 1 067 numeric values, 267 source text assignments, 2 756 unknowns and 264 inapplicable entries. Broad electronic configuration, ionization, affinity, level and line tables remain unacquired; their unknown entries do not assert that physical measurements are unavailable. Neutral-atom masses, decay channels and state assignments apply to the curated isotope set of §9, extended by the pilot identities.

Standard atomic weight and isotope abundance describe a stated material or terrestrial composition convention. They remain separate from the mass of an individual isotope. Mass excess, binding energy, separation energy and decay energy retain their source definitions and electron conventions. A derived difference requires compatible input masses and a covariance treatment before it can acquire an uncertainty.

For the retained pilot histories, write q_i for each generic bounded cadence in its configuration-specific window. One deliberately hypothetical cross-configuration bridge is

$$\widetilde{\nu}_i = \nu_{\mathrm{Cs}} \frac{q_i}{q_{\mathrm{Cs}}} \left(1 + \lambda \frac{55 - Z_i}{54}\right), \qquad \lambda \geq 0. \tag{40}$$

Here ν_{Cs} is the retained defining frequency. This choice normalizes Cs for every λ but does not identify the generic cadence with an atomic transition. The executed hydrogen-isotope cadences are $q_{\mathrm{H}} = 1/3$, $q_{\mathrm{D}} = 1/22$ and $q_{\mathrm{T}} = 1/35$. Since all three have $Z = 1$, their shared factor cancels in D/H and T/H, fixing those ratios throughout this stipulated family.

Calibration. The existing hydrogen and caesium records retain their exposure and roles from §7. A defining frequency is exact as a convention; realization uncertainty belongs to the realization. Already seen spectroscopic targets remain retrospective. A frozen model protocol fixes its inputs, common calibration and comparison domain before the separate target join. Missing state or observation maps leave the corresponding prediction undefined; they do not produce a zero residual.

Falsification. A charge mismatch, substituted isotope, changed transition endpoint or removed source qualification invalidates a comparison. A new target value must not alter the generated native history or prediction inputs. Matched feedback and passive-observation controls distinguish a dynamical intervention from a display operation. The shared finite dynamics and its state correspondence must support every claimed observable without an independently tuned gain for each property. In the stipulated pilot bridge family, the common gain cancels between hydrogen isotopes. The resulting D/H and

T/H ratios are respectively 3/22 and 3/35, outside the declared retrospective envelopes formed from the retained quoted uncertainty magnitudes. This rejects that bridge family under its stated diagnostic criterion. Those envelopes are not a joint confidence region; the result supplies neither a completed atomic observation law nor a rejection of other AOD models.

Provenance. The additive catalog and coverage matrix reside in `manual2/data/m1/properties/`. Its model protocol and separately generated comparisons retain the distinction between a native construction, an application choice and an empirical observable. The original element ledger and its 126 reference positions remain those of App. D.

Literature note. The offline atomic adoption interface addresses the NIST Atomic Spectra Database[15]. Acquired neutral-atom masses use AME2020[22]; spin, abundance and decay assignments use NUBASE2020[8], with dated moment and radius records from ENSDF[16]. The retained source manifests specify the acquired files, locators and qualifications for each adopted value.

12 Calculation and Data Registry

Setup. Each record identifies its data support before its report coordinate. D0 denotes an exact internal fixture comparison; O0 denotes a declared observable-map comparison. A dash means that this row is a reference, source or acquisition record rather than such a comparison. Record form, model premises and execution evidence retain the separate meanings of §1.1. The registry locates calculations and sources; it is not an agreement verdict. The application roles below are descriptive. Normative record forms and claim classes are bound component by component in `manual2/data/m1/classification/data_cards.json`.

Table 15: **Matter I calculation and data forms.** Support applies to the specified comparison, not to every field in its source artifact.

Record	Application role	Support	Input and report coordinate
Support polynomial	Reference calculation	—	Exact integer support expression
Enclosure groups and local tuples	Reference calculation	—	Group membership and local role projection
Reference-A and assigned classes	Input manifest	—	Pinned selections and assigned inputs; original selection provenance retained
Conditional support and residuals	Reference calculation	—	Consequences of the assigned inputs, distinct from admitted accessors
v94 placement crosswalk	Historical comparison	—	118 rows and six fields; 708 placement comparisons
v94 observable inventory	Benchmark manifest	—	118 elements across E0–E5; original declaration scope
r24 occurrence/packet crosswalk	Historical declaration	—	Occurrence, packet and SADAR identifiers
Element-ledger D.E.C. declaration	Benchmark manifest	—	Required boundary, edge, kernel, return and certificate fields
Two-face construction	Exact geometric example	—	Injective Q4 incidence and illustrative traversal counts
Finite fold and recurrence v1	Declared model	—	Mapped histories, rational cadence/lag and assigned contact-duration map
Sequential recurrence v2	Declared model	—	Matched capacity intervention, later retention, returns and contacts

Record	Application role	Support	Input and report coordinate
SI observation layer	Metrology record	—	Synthetic observation maps, interval coordinate and joint uncertainty examples
BIPM adjustment data	Measured-source reduction	—	Published frequency/covariance records and 105 rounding-aware checks
Cs/H diagnostic comparison	Observable-map record	O0	Unselected gain countermodels and retrospective target residuals
Conventional hyperfine benchmark	External model record	—	Imported splitting and spin-Hamiltonian numerical response
Hydrogen/caesium inventory	Measured target / calibration anchor	—	Two isotopes, four states, two transitions and derived reference ratio
Finite isotope targets	Measured target record	—	Evaluated half-lives, uncertainties, tentative state assignments and channels
Nonpoint isotope inventory	Benchmark manifest	—	Stability qualifications, unknown values, unobserved declarations and rejected candidates
Survival ensemble	Declared model	—	Exact finite history law; median in reference-return counts
M3 mapping and BIPM precision	Metrology / exact arithmetic record	—	Typed finite-law moments and conditional printed-zero precision check
M3 metrology applications	Observable-map / source-reduction record	O0	R03 diagnostic maps, H/Cs source reductions and retrospective residuals
All-element inner recurrence	Declared model	—	135 generic inventories; 145 source/state records; bounded cadence and lag
Signed current pair audits	Exact fixture	D0	Integer contact responses and opposite views; zero pair residual
H-2/H-3 target supplement	Measured target / source acquisition	—	Hydrogen-referenced D frequency and evaluated T rescaling; original T numerical record untranscribed
Selected Lv events	Measured target record	—	Two sourced implantation-to-decay intervals, distinct from half-life
Other mass and spectral records	Benchmark manifest	—	Required quantity, source, report map and comparison coordinate

Specialization. The original v94 observable matrix has 708 rows: 118 E0 identity/registry rows, and 118 rows in each of E1 placement/classification, E2 atomic mass/standard atomic weight, E3 isotope support, E4 spectra and E5 reaction/routes. E0 is the active historical registry; E1–E5 are unscored declarations. Its ten value maps were not attached. That historical scope is separate from the six-field placement comparison, and the exact predecessor status tokens remain in the source crosswalk.

The r24 crosswalk retains occurrence, transition-packet and SADAR declaration IDs as historical identifiers. These declarations contain no executed source-to-destination event witnesses or measured scores. Correspondence of the original and public atlas is bound by parsed manifest content and five matching file hashes; it is not an identity claim about an entire public repository.

Falsification. A target record without its value map and compatible coordinate cannot acquire a residual. A source hash cannot supply a construction witness. A historical declaration cannot substitute for an executed transition. These distinctions apply independently of the D0 or O0 support code.

Provenance. The release provenance records the reviewed PR, original v94/r24 archive hashes, extracted member hashes, the six-field mapping, historical identifier joins and the current input bindings. Small verified source extracts and the generated `declaration_crosswalk.csv`, `observable_coverage.csv` and `value_map_status.csv` are stored under `manual2/data/m1/history/`. Their bindings are in `manual2/evidence/historical_sources.json`; current input hashes are in `manual2/evidence/input_manifest.json`. Exact static checks, source identity checks and any future scientific claims retain separate result scopes.

A Exact Arithmetic Fixtures

A.1 Scope

These are arithmetic fixtures. They do not claim simulated native trajectories or certified closure. The inherited four-step convention is in Main App. I; polynomial support is in Main App. H.

A.2 Four-step decompositions

For $L = 4c_L + r_L$, $r_L \in \{0, 1, 2, 3\}$, the outer even indices give

$$\begin{aligned} 6 &= 4(1) + 2, & 8 &= 4(2) + 0, & 10 &= 4(2) + 2, \\ 12 &= 4(3) + 0, & 14 &= 4(3) + 2, & 16 &= 4(4) + 0, \\ 18 &= 4(4) + 2, & 20 &= 4(5) + 0, & 22 &= 4(5) + 2. \end{aligned}$$

Even L has remainder 0 or 2; an odd index has remainder 1 or 3. This division identity is not a theorem that every closure has four steps. The inherited history-weighted load requires its declared histories and weights (Main (177)).

A.3 Labelled-reference occupancy

The 126 declared slots and 118 element labels give

$$\frac{118}{126} = \frac{59}{63}, \qquad \frac{N_{\mathrm{labelled}}(\mathsf{C}_{22})}{22} = \frac{14}{22} = \frac{7}{11}, \qquad \frac{N_{\mathrm{unfilled}}(\mathsf{C}_{22})}{22} = \frac{8}{22} = \frac{4}{11}.$$

These are exact occupancy fractions, independent of assigned TTL values and native retention.

A.4 Falsification

Non-integer quotient/remainder, an invalid remainder, or disagreement with the labelled-slot counts fails the corresponding arithmetic fixture. It does not by itself falsify a native execution that has not been performed.

Provenance. Main App. I.2, Main App. I.4, (11).

B Quarantined External Comparison

B.1 Target records and numeric policy

Declare each external record before its comparison map. A measured target identifies the quantity, isotope and state where applicable, acquisition coordinate, value, unit, uncertainty and source locator. An exact defining value has the separate form of a calibration anchor. Limits, observational stability and unmeasured quantities retain their own meanings; they are not ordinary point targets.

The hydrogen record in Table 6 is a dated measured transition frequency with reported one-sigma uncertainty; the caesium record is the defining frequency anchor. Its physical realization has separate uncertainty records. The isotope tables of §9 carry evaluated total or partial half-lives, source qualifications and nuclear state assignments. The Lv records of §9.2 are individual implantation-to-decay delays. These quantities have different measurement protocols and report coordinates. The D/T frequencies of §7.2 carry a shared hydrogen-reference dependence. Their derived-external coordinates are distinct from an independent absolute-frequency experiment.

For each displayed measured decimal retain

$$(x_{\mathrm{disp}}, u_x, \mathrm{unit}(x), \mathrm{src}(x), \mathrm{round}(x), \mathrm{scope}(x)). \tag{41}$$

The Manual display scopes are internal, external, measured, derived external, value map and demonstration. They identify the displayed value's role; they do not turn a decimal transcription into an exact physical measurement. The integer reference A and the count pair $(Z, A - Z)$ retain their separate bookkeeping roles.

B.2 Observable-map comparisons

An O0 record binds a native or declared-model candidate, its observation/value map and a measured target record before evaluating

$$\mathsf{Q}_{\mathrm{ext}}(\nu_{Z,A\leftrightarrow\Omega}; \mathcal{T}_Z). \tag{42}$$

The candidate and inputs are fixed before the evaluation target is joined. The map supplies a common quantity, coordinate and applicable window. Residuals belong to that coordinate; a standardized residual additionally requires the applicable uncertainty and dependence model. External constants and measured values enter this reporting layer, not the native D.E.C. calculation.

The hydrogen/caesium reference ratio is a derived external value. The corrected Table IV and Table VI hydrogen values in §8.4 are source reductions. The observation-family residuals in §8 are conditional countermodel comparisons with an exposed target. None supplies the atomic observation law: the finite histories and caesium normalization admit multiple hydrogen reports. Likewise, the bounded campaign return ratios of §10 require a state-to-emitted-phase map and a common measurement protocol before an atomic frequency comparison. An isotope half-life additionally requires a state-dependent loss observation and a realized time coordinate.

Provenance. Main §5.16.12; AOD Manual I at commit `71cbcf7`, `manual/sections/00_scope.tex`, labels `manual:record-forms` and `manual:exact-arithmetic-presentation`; `manual/appendices/B_quarantined_external_comparison.tex`, `manual:external-comparison`.

Literature note. Historical hydrogen measurement: Hellwig et al. [5]. Defining reference: BIPM [1]. Isotope evaluations: ENSDF [16] and NUBASE2020 [8]. General mass and spectral sources: Wang et al. [22]; NIST ASD [15]. These citations do not assign a provenance to the original element reference-A selections.

C Exact Parity, Role and Occupancy Comparators

C.1 Integer comparators

On each complete reference group C_{2m}, $m = 3, \ldots, 11$:

1. Exactly m seats have odd local parity and m even local parity. Against the comparator $\widehat{O}_m = m$,

$$\Delta^{\mathbb{Z}}_{\text{parity},m} = \widehat{O}_m - O_{\text{even},m} = 0.$$

   Parity does not determine the assigned operational class.

2. The declared role projection has $m-1$ first-branch seats, one hinge seat, $m-1$ counterbranch seats and one boundary seat. Thus

$$\Delta^{\mathbb{Z}}_{\text{branch},m} = (m-1) - (m-1) = 0.$$

   These are counts of visual roles, not traversed edge events.

3. The terminal group has 14 labelled slots and capacity 22:

$$\Delta^{\mathbb{Z}}_{\mathsf{C}_{22}} = 14 - 22 = -8.$$

   The difference counts eight unfilled reference slots and is not a retention or physical-cutoff residual.

C.2 Signed-magnitude record

The comparator representation is

$$\Delta^{\mathbb{Z}}_k = s^{(3)}_k m_k, \qquad s^{(3)}_k \in \{-1, 0, +1\}, \quad m_k = |\Delta^{\mathbb{Z}}_k|.$$

The two balanced comparisons give $\delta_3 = (0, 0)$; the terminal occupancy comparison gives $\delta_3 = (-1, 8)$. These values express the declared integer counts, not a native phase-cycle measurement.

Provenance. (12), (13), Main §5.7.13; released AOD Manual v41.02, App. C, for the signed comparator convention.

D Complete Element Reference Ledger

The following table preserves the full declared reference. Its numerical and historical fields are unchanged by the isotope and survival extension.

Table 16: **Complete declared reference and conditional support ledger.** $Z = 1 \ldots 118$ names supplied elements; slots $119 \ldots 126$ are unfilled reference positions. Pressure and route classes are assigned inputs; the support sum is a conditional expression, not an admitted native duration. Grouping follows Main §5.7.13; the expression specializes Main (35).

Z	Sym	Ref.	A	C_{2m}	Local $a{:}b{:}c$	u_-	h	u_+	U	P^{fix}	C^{fix}	$S^{\text{fix}}, X^{\text{fix}}$	Assigned class
1	H	1		C_6	1 : 0 : 1	30	1	4	35	15	18	$-3, 0$	Deficit
2	He	4		C_6	1 : 1 : 0	196	1	8	205	18	18	$0, 0$	Neutral
3	Li	7		C_6	2 : 1 : 1	362	1	12	375	18	18	$0, 0$	Neutral

Z	Sym	Ref.	A	C_{2m}	Local $a{:}b{:}c$	u_-	h	u_+	U	P^{fix}	C^{fix}	$S^{\mathrm{fix}}, X^{\mathrm{fix}}$	Assigned class
4	Be	9		C_6	2 : 2 : 0	460	1	16	477	18	18	0, 0	Neutral
5	B	11		C_6	3 : 2 : 1	558	1	20	579	18	18	0, 0	Neutral
6	C	12		C_6	3 : 3 : 0	588	1	24	613	18	18	0, 0	Neutral + anchor
7	N	14		C_8	1 : 0 : 1	686	1	4	691	24	24	0, 0	Neutral
8	O	16		C_8	1 : 1 : 0	784	1	8	793	21	24	−3, 0	Deficit
9	F	19		C_8	2 : 1 : 1	950	1	12	963	21	24	−3, 0	Deficit
10	Ne	20		C_8	2 : 2 : 0	980	1	16	997	24	24	0, 0	Neutral
11	Na	23		C_8	3 : 2 : 1	1146	1	20	1167	24	24	0, 0	Neutral
12	Mg	24		C_8	3 : 3 : 0	1176	1	24	1201	24	24	0, 0	Neutral
13	Al	27		C_8	4 : 3 : 1	1342	1	28	1371	24	24	0, 0	Neutral
14	Si	28		C_8	4 : 4 : 0	1372	1	32	1405	24	24	0, 0	Neutral + anchor
15	P	31		C_{10}	1 : 0 : 1	1538	1	4	1543	30	30	0, 0	Neutral
16	S	32		C_{10}	1 : 1 : 0	1568	1	8	1577	27	30	−3, 0	Deficit
17	Cl	35		C_{10}	2 : 1 : 1	1734	1	12	1747	27	30	−3, 0	Deficit
18	Ar	40		C_{10}	2 : 2 : 0	2036	1	16	2053	30	30	0, 0	Neutral
19	K	39		C_{10}	3 : 2 : 1	1930	1	20	1951	30	30	0, 0	Neutral
20	Ca	40		C_{10}	3 : 3 : 0	1960	1	24	1985	30	30	0, 0	Neutral
21	Sc	45		C_{10}	4 : 3 : 1	2262	1	28	2291	30	30	0, 0	Neutral
22	Ti	48		C_{10}	4 : 4 : 0	2428	1	32	2461	30	30	0, 0	Neutral
23	V	51		C_{10}	5 : 4 : 1	2594	1	36	2631	30	30	0, 0	Neutral
24	Cr	52		C_{10}	5 : 5 : 0	2624	1	40	2665	30	30	0, 0	Neutral + anchor
25	Mn	55		C_{12}	1 : 0 : 1	2790	1	4	2795	36	36	0, 0	Neutral
26	Fe	56		C_{12}	1 : 1 : 0	2820	1	8	2829	36	36	0, 0	Neutral
27	Co	59		C_{12}	2 : 1 : 1	2986	1	12	2999	36	36	0, 0	Neutral
28	Ni	58		C_{12}	2 : 2 : 0	2880	1	16	2897	36	36	0, 0	Neutral
29	Cu	63		C_{12}	3 : 2 : 1	3182	1	20	3203	36	36	0, 0	Neutral
30	Zn	64		C_{12}	3 : 3 : 0	3212	1	24	3237	36	36	0, 0	Neutral
31	Ga	69		C_{12}	4 : 3 : 1	3514	1	28	3543	36	36	0, 0	Neutral
32	Ge	74		C_{12}	4 : 4 : 0	3816	1	32	3849	36	36	0, 0	Neutral
33	As	75		C_{12}	5 : 4 : 1	3846	1	36	3883	36	36	0, 0	Neutral
34	Se	80		C_{12}	5 : 5 : 0	4148	1	40	4189	33	36	−3, 0	Deficit
35	Br	79		C_{12}	6 : 5 : 1	4042	1	44	4087	33	36	−3, 0	Deficit
36	Kr	84		C_{12}	6 : 6 : 0	4344	1	48	4393	36	36	0, 0	Neutral + anchor
37	Rb	85		C_{14}	1 : 0 : 1	4374	1	4	4379	42	42	0, 0	Neutral
38	Sr	88		C_{14}	1 : 1 : 0	4540	1	8	4549	42	42	0, 0	Neutral
39	Y	89		C_{14}	2 : 1 : 1	4570	1	12	4583	42	42	0, 0	Neutral
40	Zr	90		C_{14}	2 : 2 : 0	4600	1	16	4617	42	42	0, 0	Neutral
41	Nb	93		C_{14}	3 : 2 : 1	4766	1	20	4787	42	42	0, 0	Neutral
42	Mo	98		C_{14}	3 : 3 : 0	5068	1	24	5093	42	42	0, 0	Neutral
43	Tc	98		C_{14}	4 : 3 : 1	5030	1	28	5059	45	42	3, 3	Surplus
44	Ru	102		C_{14}	4 : 4 : 0	5264	1	32	5297	42	42	0, 0	Neutral
45	Rh	103		C_{14}	5 : 4 : 1	5294	1	36	5331	42	42	0, 0	Neutral
46	Pd	106		C_{14}	5 : 5 : 0	5460	1	40	5501	42	42	0, 0	Neutral
47	Ag	107		C_{14}	6 : 5 : 1	5490	1	44	5535	42	42	0, 0	Neutral
48	Cd	114		C_{14}	6 : 6 : 0	5928	1	48	5977	42	42	0, 0	Neutral
49	In	115		C_{14}	7 : 6 : 1	5958	1	52	6011	42	42	0, 0	Neutral
50	Sn	120		C_{14}	7 : 7 : 0	6260	1	56	6317	42	42	0, 0	Neutral + anchor
51	Sb	121		C_{16}	1 : 0 : 1	6290	1	4	6295	48	48	0, 0	Neutral
52	Te	130		C_{16}	1 : 1 : 0	6864	1	8	6873	45	48	−3, 0	Deficit
53	I	127		C_{16}	2 : 1 : 1	6622	1	12	6635	45	48	−3, 0	Deficit
54	Xe	132		C_{16}	2 : 2 : 0	6924	1	16	6941	48	48	0, 0	Neutral
55	Cs	133		C_{16}	3 : 2 : 1	6954	1	20	6975	48	48	0, 0	Neutral
56	Ba	138		C_{16}	3 : 3 : 0	7256	1	24	7281	48	48	0, 0	Neutral
57	La	139		C_{16}	4 : 3 : 1	7286	1	28	7315	48	48	0, 0	Neutral
58	Ce	140		C_{16}	4 : 4 : 0	7316	1	32	7349	48	48	0, 0	Neutral
59	Pr	141		C_{16}	5 : 4 : 1	7346	1	36	7383	48	48	0, 0	Neutral
60	Nd	142		C_{16}	5 : 5 : 0	7376	1	40	7417	48	48	0, 0	Neutral
61	Pm	145		C_{16}	6 : 5 : 1	7542	1	44	7587	51	48	3, 3	Surplus
62	Sm	152		C_{16}	6 : 6 : 0	7980	1	48	8029	48	48	0, 0	Neutral
63	Eu	153		C_{16}	7 : 6 : 1	8010	1	52	8063	48	48	0, 0	Neutral
64	Gd	158		C_{16}	7 : 7 : 0	8312	1	56	8369	48	48	0, 0	Neutral
65	Tb	159		C_{16}	8 : 7 : 1	8342	1	60	8403	48	48	0, 0	Neutral
66	Dy	164		C_{16}	8 : 8 : 0	8644	1	64	8709	48	48	0, 0	Neutral + anchor
67	Ho	165		C_{18}	1 : 0 : 1	8674	1	4	8679	54	54	0, 0	Neutral
68	Er	166		C_{18}	1 : 1 : 0	8704	1	8	8713	54	54	0, 0	Neutral
69	Tm	169		C_{18}	2 : 1 : 1	8870	1	12	8883	54	54	0, 0	Neutral
70	Yb	174		C_{18}	2 : 2 : 0	9172	1	16	9189	54	54	0, 0	Neutral
71	Lu	175		C_{18}	3 : 2 : 1	9202	1	20	9223	54	54	0, 0	Neutral

Z	Sym	Ref.	A	C_{2m}	Local $a{:}b{:}c$	u_-	h	u_+	U	P^{fix}	C^{fix}	$S^{\mathrm{fix}}, X^{\mathrm{fix}}$	Assigned class
72	Hf	180		C_{18}	3 : 3 : 0	9504	1	24	9529	54	54	0, 0	Neutral
73	Ta	181		C_{18}	4 : 3 : 1	9534	1	28	9563	54	54	0, 0	Neutral
74	W	184		C_{18}	4 : 4 : 0	9700	1	32	9733	54	54	0, 0	Neutral
75	Re	187		C_{18}	5 : 4 : 1	9866	1	36	9903	54	54	0, 0	Neutral
76	Os	192		C_{18}	5 : 5 : 0	10168	1	40	10209	54	54	0, 0	Neutral
77	Ir	193		C_{18}	6 : 5 : 1	10198	1	44	10243	54	54	0, 0	Neutral
78	Pt	195		C_{18}	6 : 6 : 0	10296	1	48	10345	54	54	0, 0	Neutral
79	Au	197		C_{18}	7 : 6 : 1	10394	1	52	10447	54	54	0, 0	Neutral
80	Hg	202		C_{18}	7 : 7 : 0	10696	1	56	10753	54	54	0, 0	Neutral
81	Tl	205		C_{18}	8 : 7 : 1	10862	1	60	10923	54	54	0, 0	Neutral
82	Pb	208		C_{18}	8 : 8 : 0	11028	1	64	11093	54	54	0, 0	Neutral
83	Bi	209		C_{18}	9 : 8 : 1	11058	1	68	11127	54	54	0, 0	Neutral
84	Po	209		C_{18}	9 : 9 : 0	11020	1	72	11093	57	54	3, 3	Surplus + anchor
85	At	210		C_{20}	1 : 0 : 1	11050	1	4	11055	63	60	3, 3	Surplus
86	Rn	222		C_{20}	1 : 1 : 0	11828	1	8	11837	63	60	3, 3	Surplus
87	Fr	223		C_{20}	2 : 1 : 1	11858	1	12	11871	63	60	3, 3	Surplus
88	Ra	226		C_{20}	2 : 2 : 0	12024	1	16	12041	63	60	3, 3	Surplus
89	Ac	227		C_{20}	3 : 2 : 1	12054	1	20	12075	63	60	3, 3	Surplus
90	Th	232		C_{20}	3 : 3 : 0	12356	1	24	12381	66	60	6, 6	Surplus
91	Pa	231		C_{20}	4 : 3 : 1	12250	1	28	12279	66	60	6, 6	Surplus
92	U	238		C_{20}	4 : 4 : 0	12688	1	32	12721	66	60	6, 6	Surplus
93	Np	237		C_{20}	5 : 4 : 1	12582	1	36	12619	66	60	6, 6	Surplus
94	Pu	244		C_{20}	5 : 5 : 0	13020	1	40	13061	66	60	6, 6	Surplus
95	Am	243		C_{20}	6 : 5 : 1	12914	1	44	12959	66	60	6, 6	Surplus
96	Cm	247		C_{20}	6 : 6 : 0	13148	1	48	13197	69	60	9, 9	Surplus
97	Bk	247		C_{20}	7 : 6 : 1	13110	1	52	13163	69	60	9, 9	Surplus
98	Cf	251		C_{20}	7 : 7 : 0	13344	1	56	13401	69	60	9, 9	Surplus
99	Es	252		C_{20}	8 : 7 : 1	13374	1	60	13435	69	60	9, 9	Surplus
100	Fm	257		C_{20}	8 : 8 : 0	13676	1	64	13741	69	60	9, 9	Surplus
101	Md	258		C_{20}	9 : 8 : 1	13706	1	68	13775	69	60	9, 9	Surplus
102	No	259		C_{20}	9 : 9 : 0	13736	1	72	13809	72	60	12, 12	Surplus
103	Lr	266		C_{20}	10 : 9 : 1	14174	1	76	14251	72	60	12, 12	Surplus
104	Rf	267		C_{20}	10 : 10 : 0	14204	1	80	14285	72	60	12, 12	Surplus + anchor
105	Db	268		C_{22}	1 : 0 : 1	14234	1	4	14239	78	66	12, 12	Surplus
106	Sg	269		C_{22}	1 : 1 : 0	14264	1	8	14273	78	66	12, 12	Surplus
107	Bh	270		C_{22}	2 : 1 : 1	14294	1	12	14307	78	66	12, 12	Surplus
108	Hs	277		C_{22}	2 : 2 : 0	14732	1	16	14749	81	66	15, 15	Surplus
109	Mt	278		C_{22}	3 : 2 : 1	14762	1	20	14783	81	66	15, 15	Surplus
110	Ds	281		C_{22}	3 : 3 : 0	14928	1	24	14953	81	66	15, 15	Surplus
111	Rg	282		C_{22}	4 : 3 : 1	14958	1	28	14987	81	66	15, 15	Surplus
112	Cn	285		C_{22}	4 : 4 : 0	15124	1	32	15157	81	66	15, 15	Surplus
113	Nh	286		C_{22}	5 : 4 : 1	15154	1	36	15191	81	66	15, 15	Surplus
114	Fl	289		C_{22}	5 : 5 : 0	15320	1	40	15361	84	66	18, 18	Surplus
115	Mc	290		C_{22}	6 : 5 : 1	15350	1	44	15395	84	66	18, 18	Surplus
116	Lv	293		C_{22}	6 : 6 : 0	15516	1	48	15565	84	66	18, 18	Surplus
117	Ts	294		C_{22}	7 : 6 : 1	15546	1	52	15599	84	66	18, 18	Surplus
118	Og	294		C_{22}	7 : 7 : 0	15508	1	56	15565	84	66	18, 18	Surplus
119	–	297		C_{22}	8 : 7 : 1	15674	1	60	15735	69	66	3, 3	Unfilled fixture
120	–	298		C_{22}	8 : 8 : 0	15704	1	64	15769	72	66	6, 6	Unfilled fixture
121	–	301		C_{22}	9 : 8 : 1	15870	1	68	15939	75	66	9, 9	Unfilled fixture
122	–	302		C_{22}	9 : 9 : 0	15900	1	72	15973	78	66	12, 12	Unfilled fixture
123	–	305		C_{22}	10 : 9 : 1	16066	1	76	16143	81	66	15, 15	Unfilled fixture
124	–	306		C_{22}	10 : 10 : 0	16096	1	80	16177	84	66	18, 18	Unfilled fixture
125	–	309		C_{22}	11 : 10 : 1	16262	1	84	16347	87	66	21, 21	Unfilled fixture
126	–	310		C_{22}	11 : 11 : 0	16292	1	88	16381	90	66	24, 24	Unfilled fixture

References

[1] Bureau International des Poids et Mesures. Si base unit: second. Accessed 26 October 2026. `https://www.bipm.org/en/si-base-units/second`.

[2] Consultative Committee for Time and Frequency. Mise en pratique for the definition of the second in the si, 2019. BIPM, SI Brochure Appendix 2. `https://www.bipm.org/documents/20126/`

41489667/SI-App2-second.pdf/3c76fec8-04d9-f484-5c3c-a2e280a0f248.

[3] J. M. Gates et al. Towards the discovery of new elements: Production of livermorium (Z=116) with ^{50}Ti, 2024. Retained source: arXiv:2407.16079v1, 22 July 2024, Fig. 2. https://arxiv.org/abs/2407.16079v1.

[4] Thomas P. Heavner, Elizabeth A. Donley, Filippo Levi, Giovanni Costanzo, Thomas E. Parker, Jon H. Shirley, Neil Ashby, Stephan Barlow, and S. R. Jefferts. First accuracy evaluation of NIST-F2. *Metrologia*, 51(3):174–182, 2014. August 2013 evaluation and typical Table 1 budget. https://doi.org/10.1088/0026-1394/51/3/174.

[5] Helmut Hellwig, Robert F. C. Vessot, Martin W. Levine, Paul W. Zitzewitz, David W. Allan, and David J. Glaze. Measurement of the unperturbed hydrogen hyperfine transition frequency. *IEEE Transactions on Instrumentation and Measurement*, IM-19(4):200–209, 1970. Printed p. 200: one-sigma uncertainty 0.002 Hz. https://tf.nist.gov/general/pdf/13.pdf.

[6] Joint Committee for Guides in Metrology. Evaluation of measurement data: Guide to the expression of uncertainty in measurement, 2008. JCGM 100:2008. https://www.bipm.org/documents/20126/2071204/JCGM_100_2008_E.pdf.

[7] Joint Committee for Guides in Metrology. Guide to the expression of uncertainty in measurement: Part 6, developing and using measurement models, 2020. JCGM GUM-6:2020. https://www.bipm.org/documents/20126/2071204/JCGM_GUM_6_2020.pdf.

[8] F. G. Kondev, M. Wang, W. J. Huang, S. Naimi, and G. Audi. The nubase2020 evaluation of nuclear physics properties. *Chinese Physics C*, 45:030001, 2021. https://www-nds.iaea.org/amdc/ame2020/nubase_4.mas20.txt.

[9] A. E. Kramida. A critical compilation of experimental data on spectral lines and energy levels of hydrogen, deuterium, and tritium. *Atomic Data and Nuclear Data Tables*, 96:586–644, 2010.

[10] H. S. Margolis, G. Panfilo, F. Collini, M. Pizzocaro, A. D. Ludlow, R. Siadat, T. Ido, and S. Bize. Data associated with the 2025 adjustment of standard frequencies, 2026. Raw input, output, covariance, README and analysis notes. https://www.bipm.org/en/doi/10.59161/StdFreq2026.

[11] B. S. Mathur, S. B. Crampton, D. Kleppner, and N. F. Ramsey. Hyperfine separation of tritium. *Physical Review*, 158(1):14–17, 1967.

[12] National Institute of Standards and Technology. Beams of atoms: The first atomic clocks. Accessed 26 October 2026. https://www.nist.gov/atomic-clocks/how-atomic-clocks-work/beams-atoms-first-atomic-clocks.

[13] National Institute of Standards and Technology. Bohr magneton in hz/t: 2022 codata recommended value. Accessed 26 October 2026. https://physics.nist.gov/cuu/Constants/Table/allascii.txt.

[14] National Institute of Standards and Technology. Handbook of basic atomic spectroscopic data: Atomic data for hydrogen. Accessed 26 October 2026. https://physics.nist.gov/PhysRefData/Handbook/Tables/hydrogentable1_a.htm.

[15] National Institute of Standards and Technology. Atomic spectra database (ASD) and ground levels and ionization energies for the neutral atoms, 2025. NIST Physical Measurement Laboratory, https://physics.nist.gov/PhysRefData/ASD/.

[16] National Nuclear Data Center. Evaluated nuclear structure data file: Adopted levels and experimental updates, 2026. Source snapshot and exact record locators in the Matter I isotope catalog; retrieved 1 November 2026. https://www.nndc.bnl.gov/ensdf/.

[17] Reginald Poyau. Axiomatic fundamentalism (AF): A logical protocol for traceable research (all versions), 2025. All versions. DOI: https://doi.org/10.5281/zenodo.17561186.

[18] Reginald Poyau. Axiomatic fundamentalism calculus (AFC), v2.4, 2025. Authority binding: v2.4, source commit 21e66cd46b9a69a1996ed4fcecda53d831b414f8. Concept DOI: https://doi.org/10.5281/zenodo.17795590.

[19] Reginald Poyau. Alpha↔omega dynamics (AΩD): The hidden temporal dynamics of stokes; the art of the leprechaun: Fractal calculus – $\mathfrak{S}$ (v41.02, main and manual), 2026. Released v41.02; source commit 6ac87ad518652249fed67af491ea4aab9d7225ce. DOI: https://doi.org/10.5281/zenodo.23426642.

[20] J. E. Sansonetti and W. C. Martin. Handbook of basic atomic spectroscopic data. *Journal of Physical and Chemical Reference Data*, 34(4), 2005. Caesium data, printed p. 1652. https://www.nist.gov/document/jpcrd690pdf.

[21] Daniel A. Steck. Cesium d line data, 2025. Revision 2.3.4, section 3. https://steck.us/alkalidata/cesiumnumbers.pdf.

[22] M. Wang, W. J. Huang, F. G. Kondev, G. Audi, and S. Naimi. The AME 2020 atomic mass evaluation (II). *Chinese Physics C*, 45:030003, 2021.

[23] M. Wang, W. J. Huang, F. G. Kondev, G. Audi, and S. Naimi. The ame 2020 atomic mass evaluation (ii). tables, graphs and references. *Chinese Physics C*, 45:030003, 2021. https://www-nds.iaea.org/amdc/ame2020/mass_1.mas20.txt.

[24] D. J. Wineland and N. F. Ramsey. Atomic deuterium maser. *Physical Review A*, 5(2):821–837, 1972.